UNITED INSTITUTE OF TECHNOLOGY, PRAYAGRAJ
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

DATA STRUCTURE (BCS301)

B. TECH. IIND YEAR

ODD SEMESTER 2025-26

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Unit-wise Syllabus

- **UNIT-1:** Introduction of Basic Terminology, Data Organization, Single and Multidimensional Arrays, Linked List (Singly, Doubly and, Circular Linked List).
- **UNIT-2:** Stack with Push & Pop, Iteration and Recursion and Queues.
- **UNIT-3:** Concept of Searching, Hashing & Collision resolution Techniques and Sorting.
- **UNIT-4:** Basic terminology of tree with their Representation.
- **UNIT-5:** Terminology used with Graph and their Representations.

UNIT-1

Detailed Syllabus

Introduction: Basic Terminology, Elementary Data Organization, Built in Data Types in C. Algorithm, Efficiency of an Algorithm, Time and Space Complexity, Asymptotic notations: Big Oh, Big Theta and Big Omega, Time-Space trade-off. Abstract Data Types (ADT).

Arrays: Definition, Single and Multidimensional Arrays, Representation of Arrays: Row Major Order, and Column Major Order, Derivation of Index Formulae for 1-D,2-D,3-D and n-D Array Application of arrays, Sparse Matrices and their representations.

Linked lists: Array Implementation and Pointer Implementation of Singly Linked Lists, Doubly Linked List, Circularly Linked List, Operations on a Linked List. Insertion, Deletion, Traversal, Polynomial Representation and Addition Subtraction & Multiplications of Single variable & Two variables Polynomial.

University Exam Questions

SECTION A

(Each question carries 2 marks.)

1. Define the term Data Structure. List some linear and non-linear data structures stating the application area where they will be used. [2017-2018]

2. Differentiate between overflow and underflow condition in a linked list. [2018-2019]

3. How can you represent a sparse matrix in memory? [2019-2020]

4. List the various operations are linked list. [2019-2020]

5. Define time space trade off. [2020-2021]

6. Differentiate array and linked list. [2020-2021

7. List the advantages of doubly linked list over single linked list.[2021-2022]

8. Define best case, average case and worst case for analyzing the complexity of a program. [2022-23]

9. What are the advantages and disadvantages of array over linked list? [2022-2023]

10. What are the various asymptotic notations? [2023-2024]

11. Explain why do we need a pointer to maintain a linked list structure? [2024-25]

SECTION B

(Each question carries 7 marks)

1. What do you understand by time space trade off? Explain best, worst and average case analysis in this respect with an example. [2017-2018]
2. What do you understand by time space trade off? Define the various asymptotic notations. [2018-2019]

1. What are the merits and the merits of array? Give two arrays of integers in ascending order, develop an algorithm to merge these arrays to form a third array sorted in ascending order. [2019-2020]
2. Write advantages and disadvantages of linked list over arrays. Write a 'C' function creating new linear linked list by selecting alternate elements of a linear linked list.[2021-2022]
3. Write a Pseudo code that will concatenate two linked lists. Function should have two parameters, pointers to the beginning of the lists and the function should link second list at the end of the first list. [2023-2024]

SECTION C

(Each question carries 7 marks.)

1. What are the various asymptotic notations? Explain Big O notation. [2017-2018]

2. Write an algorithm to insert a node at the end in a Circular linked list. [2017-2018]

3. Write a program in C to delete a specific element in single linked list. Double linked list takes more space than single linked list for storing an extra address. Under what condition, could a double linked list more beneficial than single linked list. [2018-2019]

4. What is doubly linked list? What are its applications? Explain how an element can be deleted from doubly linked list using C program.[2019-2020]

5. Define the following terms in brief" [2019-2020]

I) Time complexity II) Space complexity III) Asymptotic notations

IV) Big O notation.

6. Discuss array and linked representation of queue data structure. What is dequeue?

7. Write a C programme to insert a node at Kth position in single linked list. [2020-2021]

8. Discuss the representation of polynomial of single variable using linked list. Write 'C' functions to add two such polynomials represented by linked list. [2021-2022] [2022-2023]

9. Discuss doubly linked list. Write an algorithm to insert a node after a given node in singly linked list. [2022-2023]

10. How will you create link list representation of a polynomial. Explain it with the suitable example. [2023-2024]

11. Write a C program for sorting 100 integer numbers using selection sort procedure. Discuss the worst-case time complexity of the algorithms. [2023-2024]

12. Illustrate the following: [2024-25]

a) Asymptotic notations b) Time Space Tradeoff c) ADT

UNIT-2

Detailed Syllabus

Stacks: Abstract Data Type, Primitive Stack operations: Push & Pop, Array and Linked Implementation of Stack in C, Application of stack: Prefix and Postfix Expressions, Evaluation of postfix expression.

Iteration and Recursion: Principles of recursion, Tail recursion, Removal of recursion Problem solving using iteration and recursion with examples such as binary search, Fibonacci numbers, and Hanoi towers. Tradeoffs between iteration and recursion.

Queues: Operations on Queue: Create, Add, Delete, Full and Empty, Circular queues, Array and linked implementation of queues in C, Dequeue and Priority Queue.

University Exam Questions

SECTION A

(Each question carries 2 marks.)

1. Convert the following arithmetic infix expression into its equivalent postfix expression. [2017-18]

2. What is recursion? give disadvantages of recursion. [2018-2019]

3.Give some applications of Stack. [2019-2020]

4. What is tail recursion? Explain with a suitable example.[2019-2020][2022-2023][2021-2022]

5. Define priority queue. Give an application of priority queue. [2019-2020]

6. Why are parentheses needed to specify the order of operations in infix expressions but not in postfix operations? [2023-2024]

7.Explain circular queue. What is the condition if circular queue is full or empty? [2017-2018] [2018-2019][2020-2021] [2022-2023] [2024-25]

SECTION B

(Each question carries 7 marks)

1. Write C function for non-recursive post order traversal. [2023-2024]
2. Write an algorithm to convert a valid arithmetic infix expression into an equivalent postfix expression. [2023-2024].
3. What is Stack? Write a C program for linked list implementation of stack. [2022-2023]

4. Write algorithm for push and pop operations in Stack. [2019-2020]

5. If a stack is implemented using Linked List, then show the logical representation of nodes as a stack. Illustrate PUSH and POP operations of this type of stack.

SECTION C

(Each question carries 7 marks.)

1. A double ended Queue (deque) is a linear list in which additions may be made at either end. Obtain a data
2. representation mapping a deque into one dimensional array, Write C function to add and delete elements from either end of deque. [2023-2024]

3. Write short notes:

   a. Differentiate between iteration and recursion. [2019-2020]

   b. Write the recursive solution for Tower of Hanoi problem. [2019-2020]4. Explain Tower of Hanoi problem and write a recursive algorithm to solve it. [2018-2019]

4. Explain how a circular queue can be implemented using arrays. Write all functions for circular queue operations.[2018-2019]

5. What is a Stack .Write a C program to reverse a string using stack. [2017-2018]

6. Define the recursion. Write a recursive and non-recursive program to calculate the factorial of the given number.[2017-2018]

7. Convert the following infix expression to reverse polish notation expression using stack. [2020-2021]

8. Write A C programme to implement stack using single linked list. [2020-2021]

9. Discuss disadvantages of recursion with some suitable example.[2021-2022]

10. Write a C program to calculate factorial of number using recursive and non-recursive functions. [2021-2022]

11. Design a method for keeping two stacks within a single linear array so that neither stack overflow until all the memory is used. [2021-2022]

12. Write a C program to reverse a string using stack. [2021-2022]

13. What is circular Queue? Write a C code to insert an element in circular queue? [2022-2023]

14. Construct a Queue using a linked list? Include functions insert(), delete(), print(). Explain how a **circular queue** overcomes the limitations of a simple queue.[2024-25]

15. Construct a C program to check whether the parentheses are balanced in an expression. [2024-25]

UNIT-3

Detailed Syllabus

Searching: Concept of Searching, Sequential search, Index Sequential Search, Binary Search. Concept of Hashing & Collision resolution Techniques used in Hashing.

Sorting: Insertion Sort, Selection, Bubble Sort, Quick Sort, Merge Sort, Heap Sort and Radix Sort.

University Exam Questions

SECTION A

(Each question carries 2 marks)

1. What do you understand by stable and in place shorting?
2. How does bubble short work? Explain.
3. Differentiate sequential search and binary search.
4. Give example of one each stable and unstable sorting techniques.
5. Illustrate the concept of Indexed Sequential search.

SECTION B

(Each question carries 7 marks)

1. How binary search is different from linear search? Apply binary search to find item 40 in the sorted array: 11,22, 30, 33,40, 44,55, 60, 66,77, 80, 88, 99. Also discuss the complexity of binary search.
2. Explain any three commonly used hash function with the suitable example.
3. Write algorithms of insertion sort. Implement the same on the following numbers; also calculate its time complexity. 13, 16, 10, 11, 4, 12, 6, 7.
4. Differentiate between liner and binary search algorithm. Write a recursive function to implement binary search.
5. Write an algorithm for Quick sort. Use Quick sort algorithm to sort the following elements: 2, 8, 7, 1, 3, 5, 6,4.

SECTION C

(Each question carries 7 marks)

1. Write short notes on: I). Heap Sort II) Hashing Technique III) Radix Sort.
2. Write algorithm for quick short. Trace your algorithm on the following data to sort the list: 2,13,4,21,7, 56,51, 85,59, 1,9,10. How the choice of pivot element effects the efficiency of algorithm.
3. What is hashing? Give the characteristics of hash function. Explain collision resolution technique in hashing.
4. Write an algorithm for merge sort and apply on following elements. 45,32, 65,76, 23,12, 54, 67, 22. 87.

5. Write a C programme for index sequential search .

6. Differentiate between linear and quadratic probing techniques.

7. Write an algorithm for Heap Sort. Use Heap sort algorithm, sort the following sequence: 18, 25, 45, 34, 36, 51, 43, and 24.

UNIT-4

Detailed Syllabus

Trees: Basic terminology used with Tree, Binary Trees, Binary Tree Representation: Array Representation and Pointer(Linked List) Representation, Binary Search Tree, Strictly Binary Tree ,Complete Binary Tree . A Extended Binary Trees, Tree Traversal algorithms: Inorder, Preorder and Postorder, Constructing Binary Tree from given Tree Traversal, Operation of Insertation, Deletion, Searching & Modification of data in Binary Search . Threaded Binary trees, Traversing Threaded Binary trees. Huffman coding using Binary Tree. Concept & Basic Operations for AVL Tree , B Tree & Binary Heaps

University Exam Questions

SECTION A

(Each question carries 2 marks)

1. Discuss the concept of "successor" and “predecessor” in Binary Search Tree.
2. Explain height balanced tree. List general cases to maintain the height.
3. Define extended binary tree, full binary tree, strictly binary tree and complete binary tree.
4. Explain threaded binary tree.
5. What is the importance of threaded binary tree?
6. Write short notes on mini heap.
7. Write advantages of AVL tree over Binary Search Tree (BST)
8. Illustrate the significance of Threaded Binary Tree.

SECTION B

(Each question carries 7 marks)

1. Define tree, binary tree, complete binary tree and full binary tree. Write algorithms or function to obtain traversals of a binary tree in preorder, postorder and inorder.
2. Explain Huffman algorithm. construct Huffman tree for MAHARASHTRA, with its optimal code.
3. What is height balanced tree? Why height by balancing of tree is required? Create an AVL tree for the following elements: a,z,b,y,c,x,d,w,e,v,f.

SECTION C

(Each question carries 7 marks)

1. Write short notes are following:

   I) Extended binary tree   II) Complete binary tree   III) Threaded binary tree

2. Write an iterative function to search a key in Binary Search Tree (BST).

3. Why does time complexity of search operation in B-Tree is better than Binary Search Tree (BST)?

4. Discuss left skewed and right skewed binary tree. Construct an AVL tree by inserting the following elements in the order of their occurrence: 60, 2, 14, 22, 13, 111, 92, 86.

5. What is B-Tree? Write the various properties of B- Tree. Show the results of inserting the keys F, S, Q, K ,C, L, H, T, V, W, M, R, N, P, A, B in order into a empty B-Tree of order 5.

6. Demonstrate B-Tree? Construct a B-Tree of order 4 with the alphabets (letters) arrive in the sequence as follows: a, g, f, b, k, d, h, m, j, e, s, I, r, x, c, l, n, t, u, p.

UNIT-5

Detailed Syllabus

Graphs: Terminology used with Graph, Data Structure for Graph Representations: Adjacency Matrices, Adjacency List, Adjacency. Graph Traversal: Depth First Search and Breadth First Search, Connected Component, Spanning Trees, Minimum Cost Spanning Trees: Prims and Kruskal algorithm. Transitive Closure and Shortest Path algorithm: Warshal Algorithm and Dijikstra Algorithm.

University Exam Questions

SECTION – A
(Each question carries 2 marks)

1. List the different types of representation of graphs.
2. How the graph can be represented in memory? Explain the suitable example.
3. What is minimum cost spanning tree? Give its applications.
4. Compare adjacency matrix and adjacency list representation of graph.
5. Writes sort notes on adjacency multi list representation graph.
6. Write different representations of graphs in the memory.
7. Write an algorithm for Breadth First Search (BFS) traversal of a graph.
8. Explain the concept of NonLinear data structures with the help of example.

SECTION B
(Each question carries 7 marks)

1. Define spanning tree. Also construct minimum spanning tree using prim’s algorithm for the given graph.

2. Write the Floyd Warshall algorithm to compute the all pairs shortest path. Apply the algorithm on following graph.

3. Write the Dijkstra algorithm for shortest path in a graph and also find the shortest path from ‘S’ to all remaining vertices of graph in the following graph.

4. Write an algorithm for breath first search item search(BFS) and explain with the help of suitable example.

5. Differentiate between DFS and BFS. Draw the breadth First Tree for the above graph.

6. Explain the Depth-First Search (DFS) algorithm with the help of an example graph. Which data structure is used for DFS and BFS?

SECTION C

(Each question carries 7 marks)

1. Describe Dijkstra’s algorithm for finding shortest path.
2. Explain the detail about the graph traversal technique with suitable examples.
3. Explain Warshall’s algorithm with the help of an example.
4. Describe the Dijkstra algorithm to find the shortest path.
5. Describe Prim’s algorithm and find the cost of minimum spanning tree using Prim’s algo..
6. Explain Floyd Warshall algorithm with the help of an example.
7. Explain two ways to represent graph in memory and compare their advantages:

   (i) Adjacency matrix (ii) Adjacency List

8. Explain the following graph terminologies with examples:
9. I) Graph II) Weighted Graph III) Degree of a Vertex

Solved Numerical
UNIT-1

Calculate the address of any element in the N-D Array:

The address LOC(C(K1, K2..... KN) of an arbitrary element of C can be obtained from the formula (column-major order)

$$LOC(C(K1, K2..... KN) = Base(C) + w[(((...\ (E_N L_{N-1} + E_{N-1})\ L_{N-2}) + E_3)L_2 + E_2)L_1 + E_1]$$

or from the formula (row-major order)

$$LOC(C(K1, K2..... KN) = Base(C) + w[(...((E_1 L_2 + E_2)L_3 + E_3)L_4 + ... + E_{N-1})\ L_N + E_N]$$

Example: Suppose a three-dimensional array MAZE is declared using MAZE(2:8, -4:1, 6:10), base address of MAZE is 200 and W=4then find the address of MAZE[5, -1, 8] using row-major order.

Solution: The lengths of the three dimensions of MAZE are, respectively.

L_1 = 8 - 2 + 1 = 7, i.e. (by the help of the formula UB-LB+1)

L_2=1+4+1=6 , L3 = 10-6+1 =5

Accordingly, MAZE contains $L_1 * L_2 * L_3$ = 7 * 6 * 5 = 210 elements.

The effective indices of the subscripts are, respectively,

E_1 = 5 - 2 = 3, E2 = -1+4 = 3, E3 = 8 - 6 = 2

Using row-major order, formula we have:

Therefore, $E_1 * L_2$ = 3*6 =18

$E_1 * L_2 + E_2$ = 18 + 3 = 21

$(E_1 * L_2 + E_2) * L_3$ = 21 * 5 = 105

$(E_1 * L_2 + E_2) * L_3 + E_3$ = 105 + 2 = 107

Therefore,

LOC(MAZE(K1, K2, K3) = Base(MAZE) + W[$((E_1 L_2 + E_2) L_3 + E_3)$]

LoC(MAZE[5, - 1, 8]) = 200 + 4*(107) = 200 + 428 = 628

Solved Numerical
UNIT-2

Example: P: 5,6,2,+,*,12,4,/,- evaluate it
Solution: Add “)” at the end of P,

S. No,	Scanned Symbol	stack
1	5	5
2	6	5,6
3	2	5,6,2
4	+	5,8
5	*	40
6	12	40,12
7	4	40,12,4
8	/	40,3
9	-	37
10	)	

Example: Convert Q: A+(B * C – (D / E ^ F) * G) * H   into postfix expression

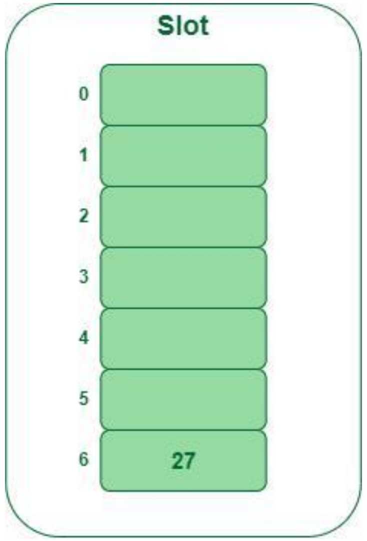
Solution: add “)” at the end of expression A+(B * C – (D / E ^ F) * G) * H) and also Push a “(“ on Stack.

Symbol Scanned	Stack	Expression(P)
	(	
A	(	A
+	(+	A
(	(+(	A
B	(+(	AB
*	(+(*	AB
C	(+(*	ABC
-	(+(-	ABC*
(	(+(-(	ABC*
D	(+(-(	ABC*D
/	(+(-(/	ABC*D
E	(+(-(/	ABC*DE
^	(+(-(/^	ABC*DE
F	(+(-(/^	ABC*DEF
)	(+(-	ABC*DEF^/
*	(+(-*	ABC*DEF^/
G	(+(-*	ABC*DEF^/G
)	(+	ABC*DEF^/G*-
*	(+*	ABC*DEF^/G*-
H	(+*	ABC*DEF^/G*-H
)	empty	ABC*DEF^/G*-H*+

Solved Numerical
UNIT-3

Example: Insert the keys 27, 43, 692, 72 into the Hash Table of size 7. where first hash-function is **h1 (k) = k mod 7** and second hash-function is **h2(k) = 1 + (k mod 5)**

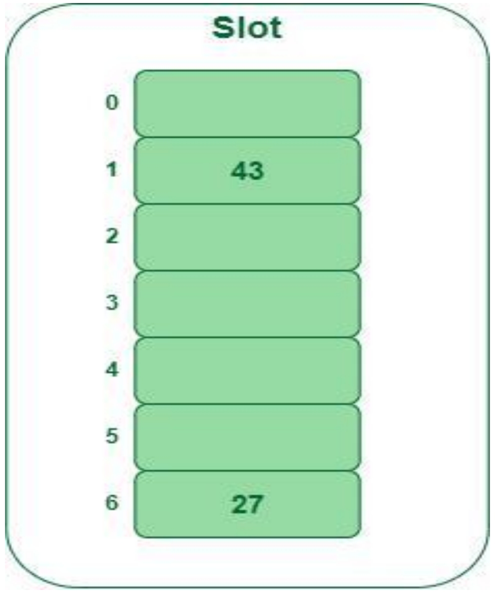
Solution: Step 1: Insert 27
27 % 7 = 6, location 6 is empty so insert 27 into 6 slot.



Insert key 27 in the hash table

Step 2: Insert 43

43 % 7 = 1, location 1 is empty so insert 43 into 1 slot.



Insert key 43 in the hash table

Step 3: Insert 692

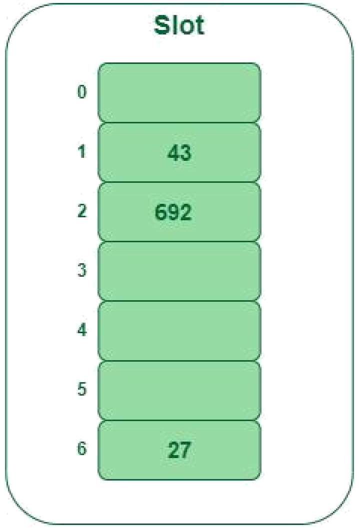
692 % 7 = 6, but location 6 is already being occupied and this is a collision, So we need to resolve this collision using double hashing.

*hnew = [h1(692) + i * (h2(692)] % 7*

[6+1(1+692%5)]%7*

9% 7 = 2

Now, as 2 is an empty slot, so we can insert 692 into 2nd slot.



Insert key 692 in the hash table

Step 4: Insert 72

72 % 7 = 2, but location 2 is already being occupied and this is a collision.

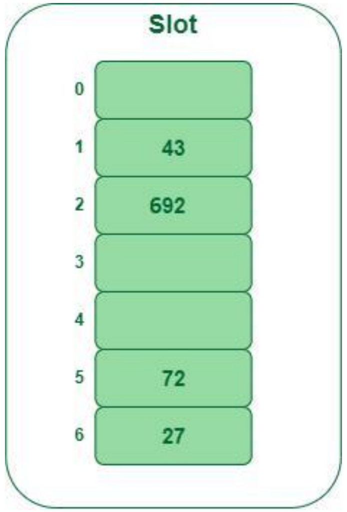
So we need to resolve this collision using double hashing. *hnew = [h1(72) + i * (h2(72)] % 7*

[2+1(1+72%5)]%7*

5 % 7

5,

Now, as 5 is an empty slot, so we can insert 72 into 5th slot.



Insert key 72 in the hash table

Solved Numerical
UNIT-4

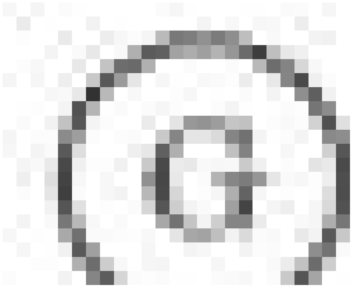
Question:. Construct a binary tree for the following :

Inorder : Q, B, K, C, F, A, G, P, E, D, H, R

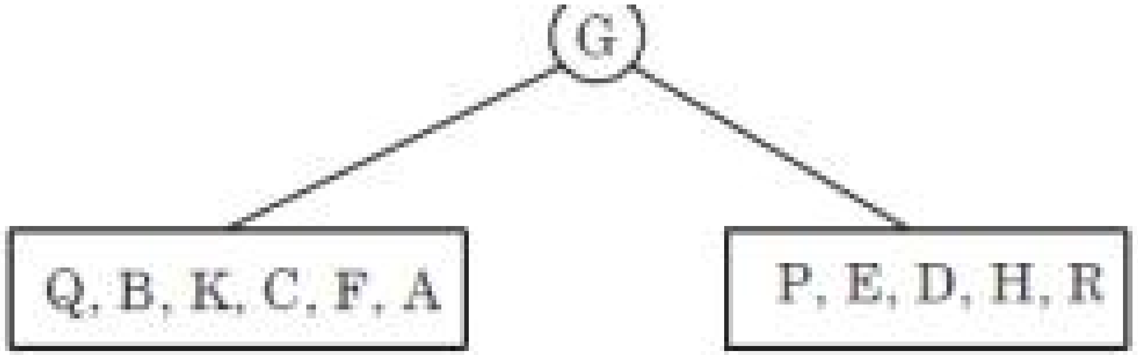
Preorder : G, B, Q, A, C, K, F, P, D, E, R, H

Find the postorder of the tree.

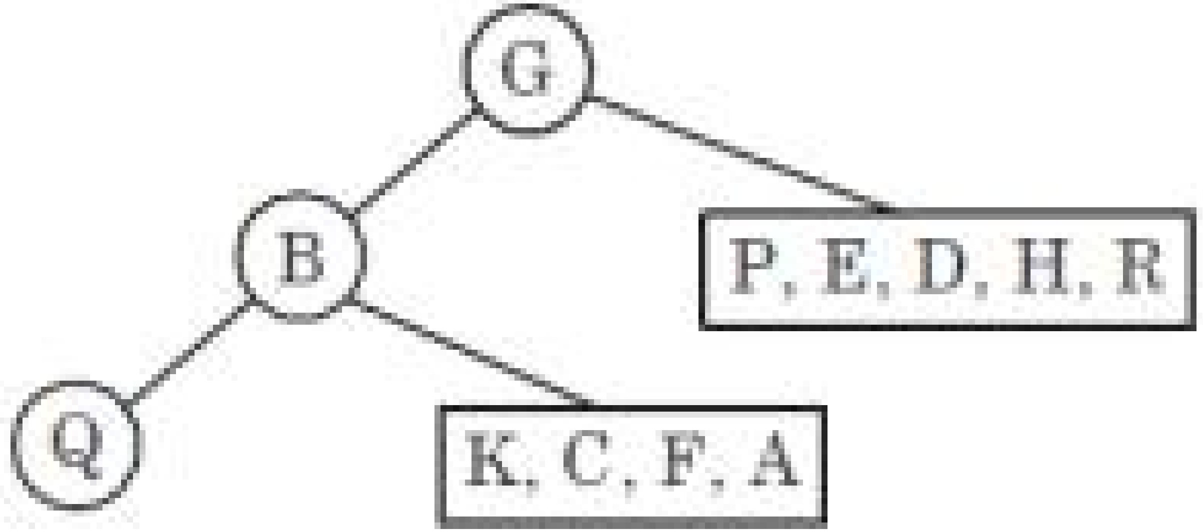
Solution: Step 1 : In preorder traversal root is the first node. So, G is the root node of the binary tree. So, G root



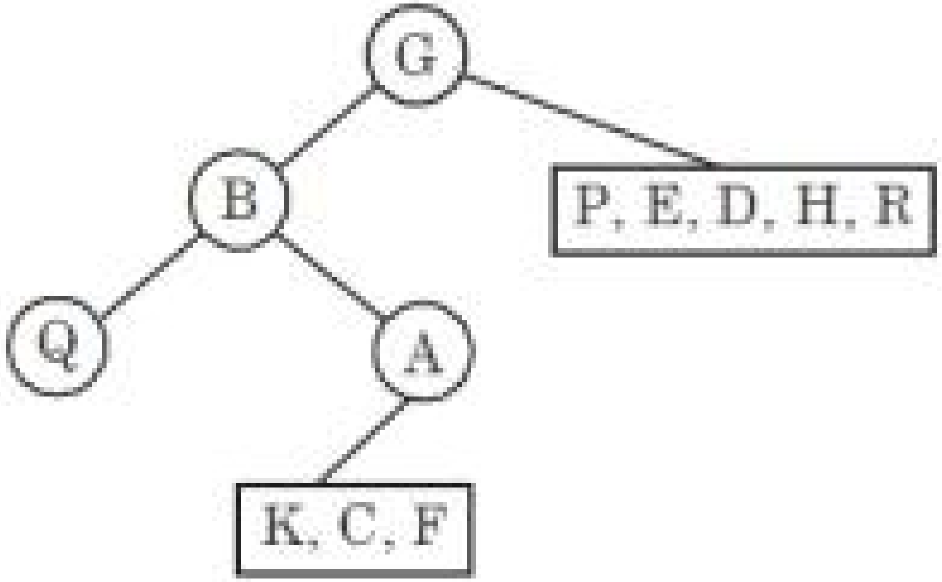
Step 2 : We can find the node of left sub-tree and right sub-tree with inorder sequence. So,



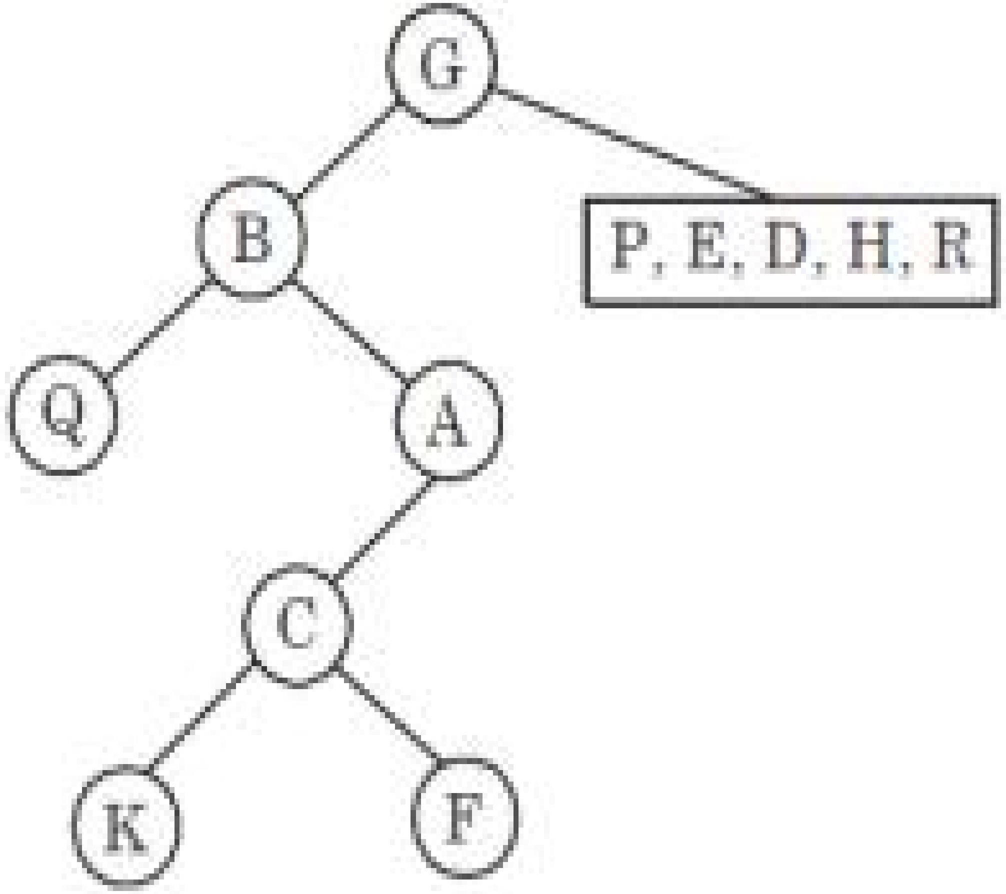
Step 3 : Now, the left child of the root node will be the first node in the preorder sequence after root node G.



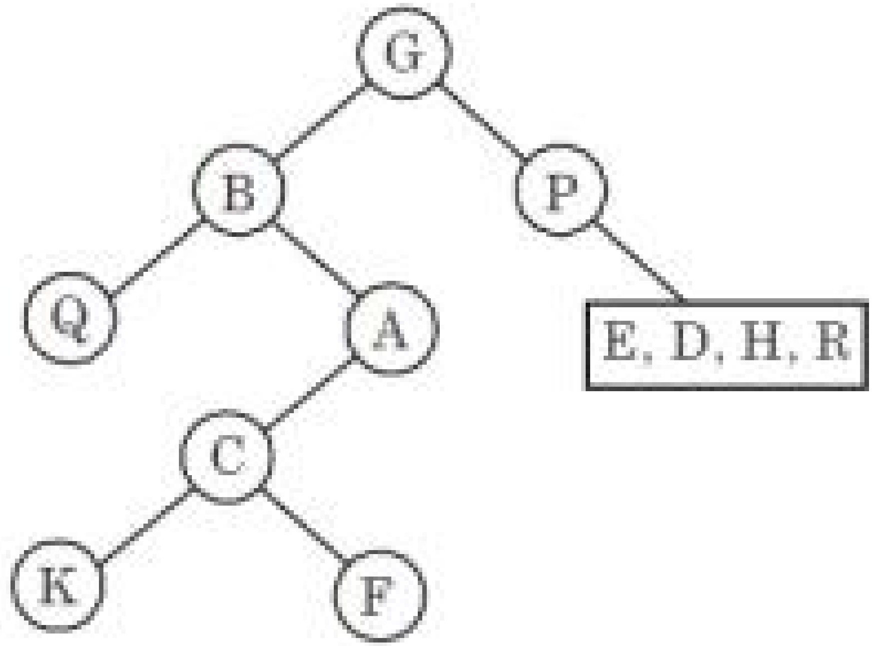
Step 4 : In inorder sequence, Q is on the left side of B and A is on the right side B. So,



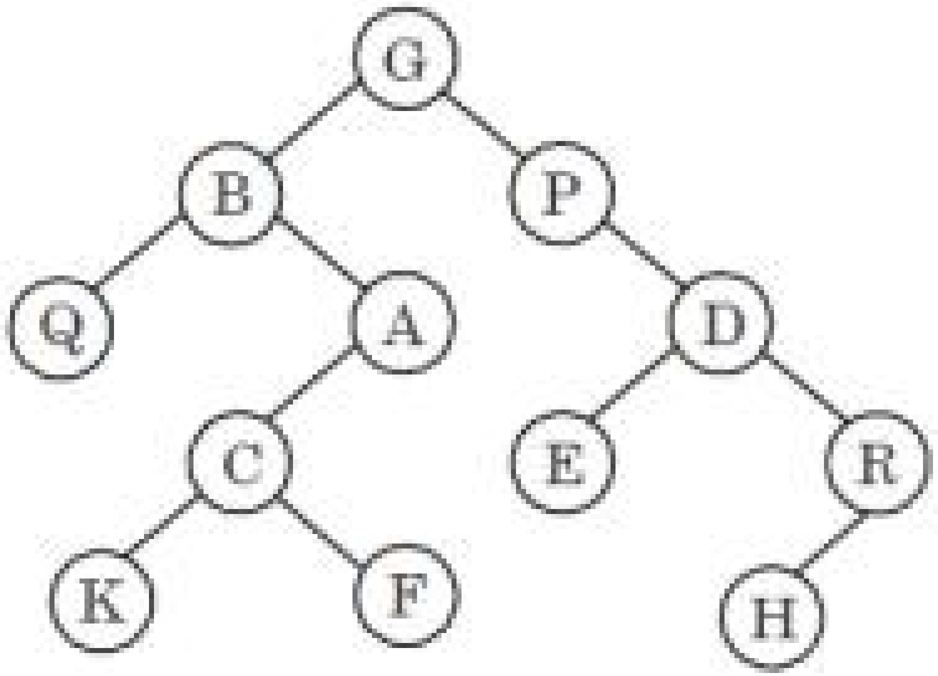
Step 5 : In inorder sequence, C is on the left side of A . Now according to inorder sequence, K is on the left side of C and F is on the right side of C.



Step 6 : Similarly, we can go further for right side of G



So, the final tree is



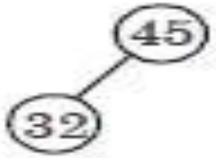
Postorder of tree : Q, K, F, A, B, E, H, R, D, P, G

Question: Make a binary search tree for the following sequence of numbers, show all steps : 45, 32, 90, 34, 68, 72, 15, 24, 30, 66, 11, 50, 10.

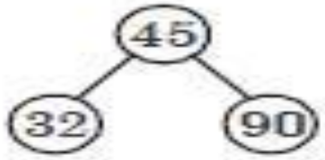
Numerical :

1. **Insert 45 :**

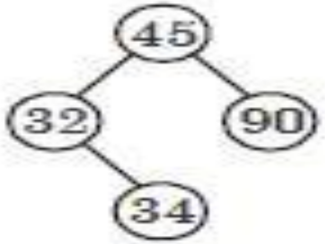
2. **Insert 32 :**



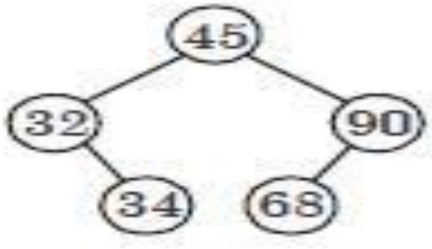
3. **Insert 90 :**



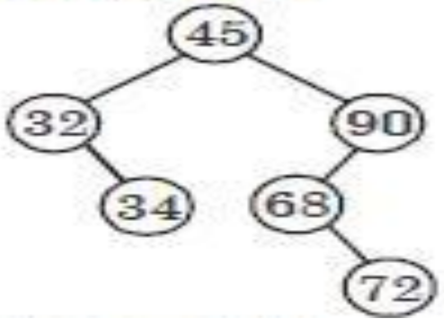
4. **Insert 34 :**



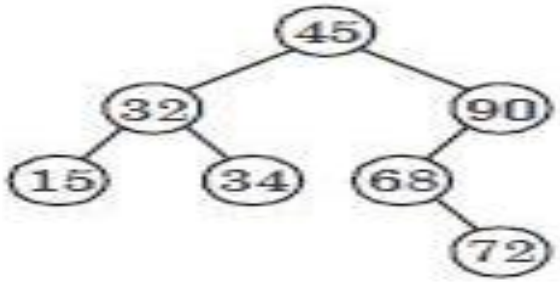
5. **Insert 68 :**



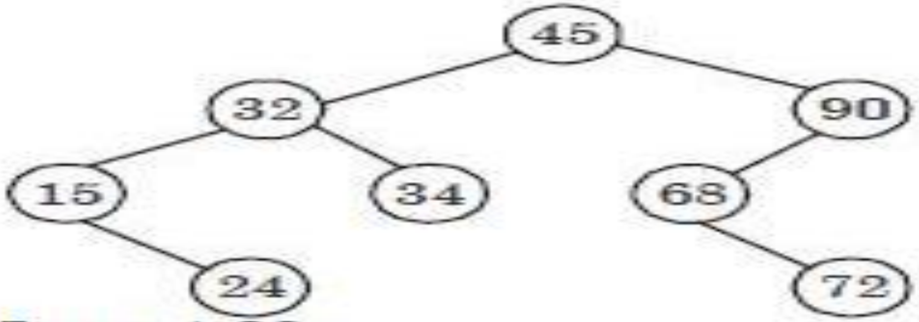
6. **Insert 72 :**



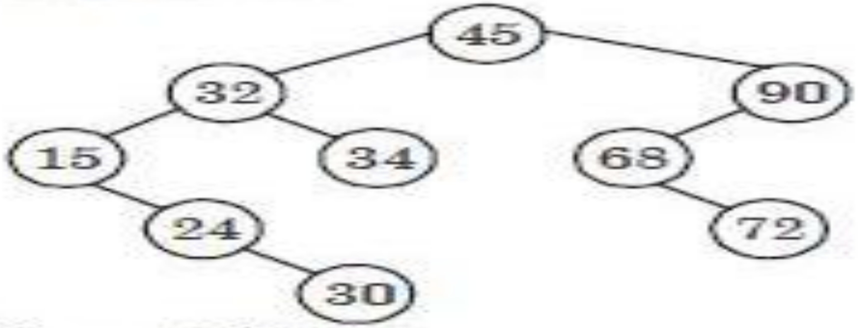
7. Insert 15 :



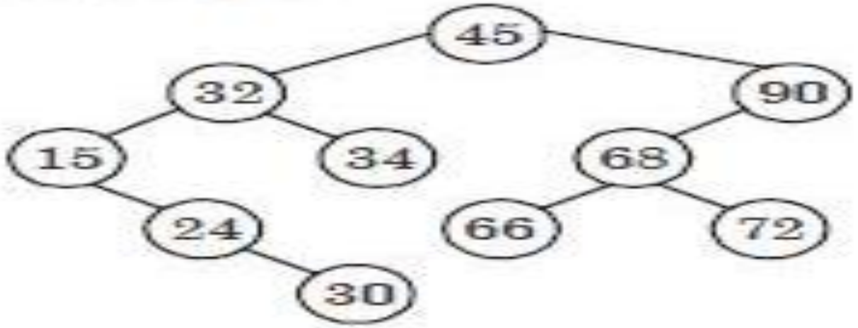
8. Insert 24 :



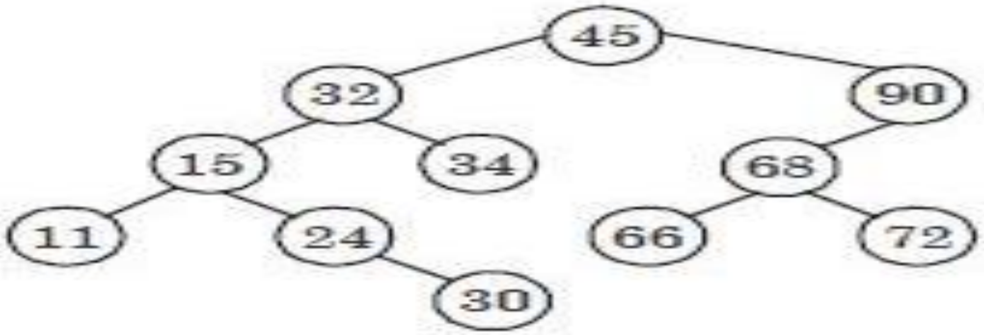
9. Insert 30 :



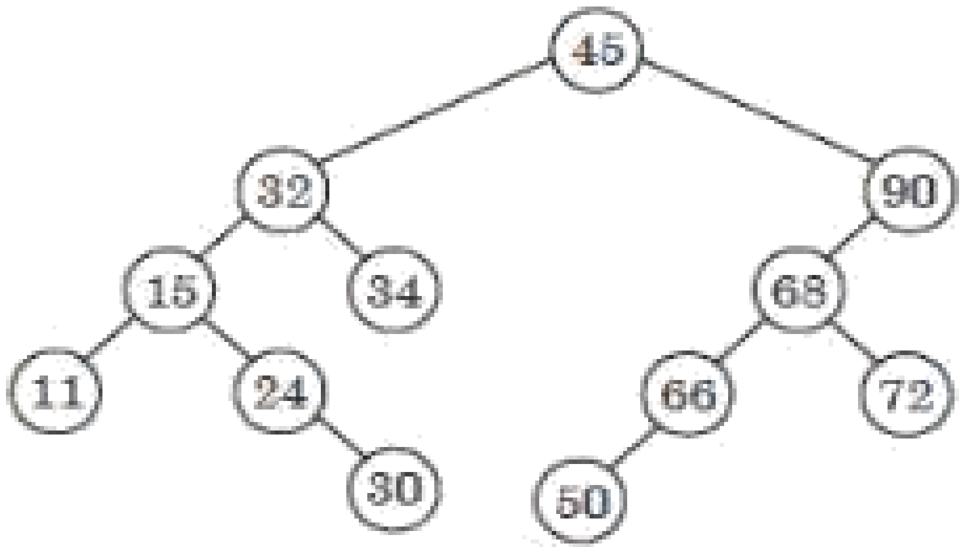
10. Insert 66 :



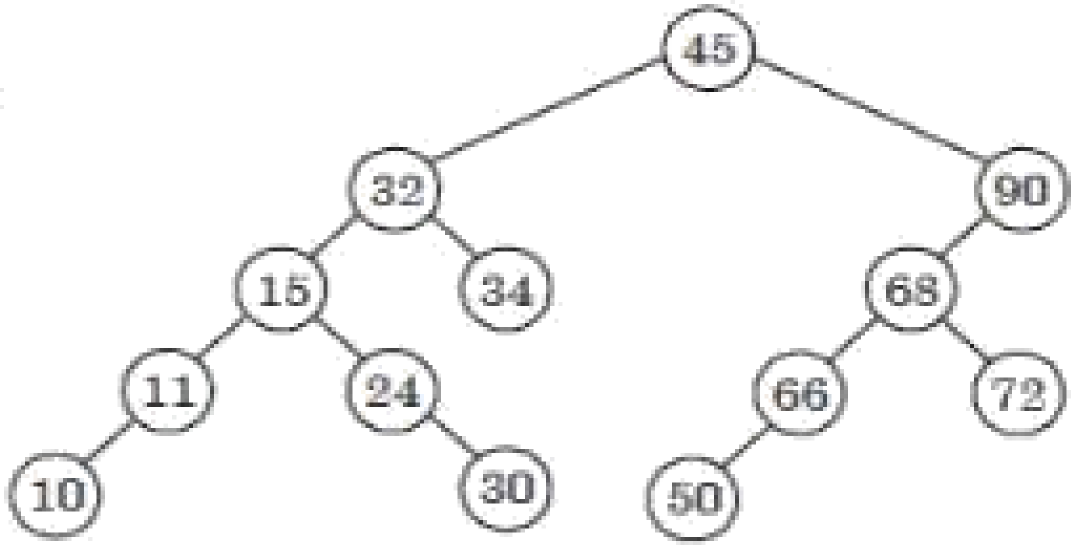
11. Insert 11 :



12. Insert 50 :

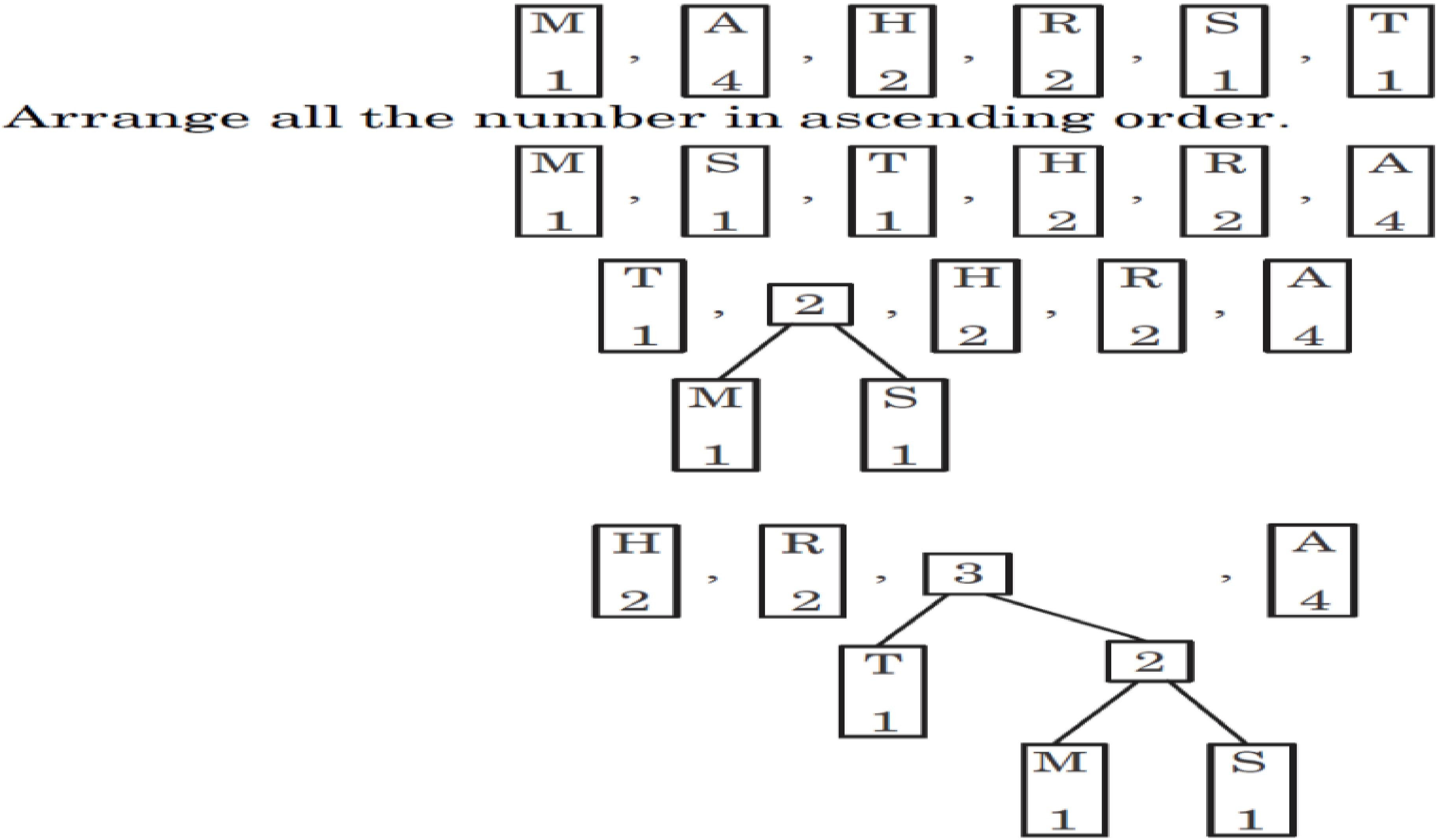


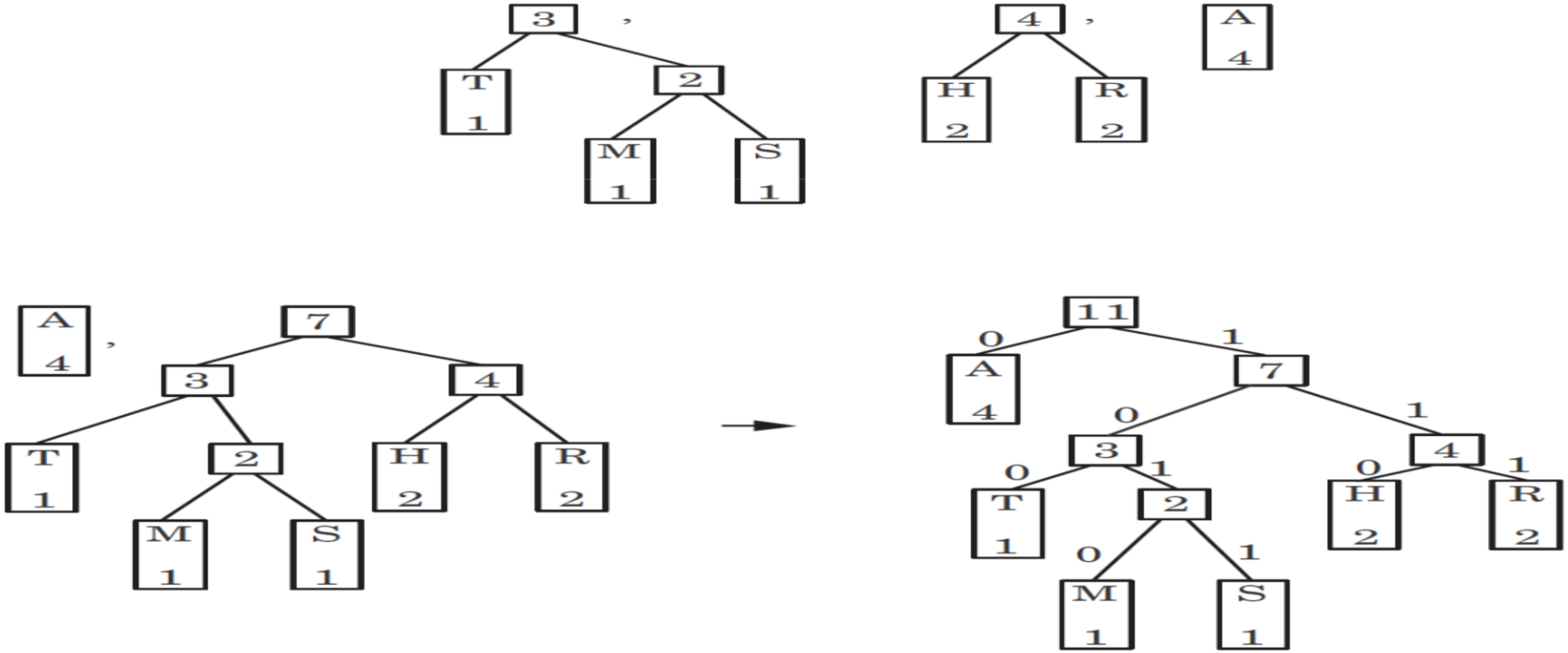
13. Insert 10 :



Question: . **Construct Huffman tree for *MAHARASHTRA* with its optimal code.**

Solution:

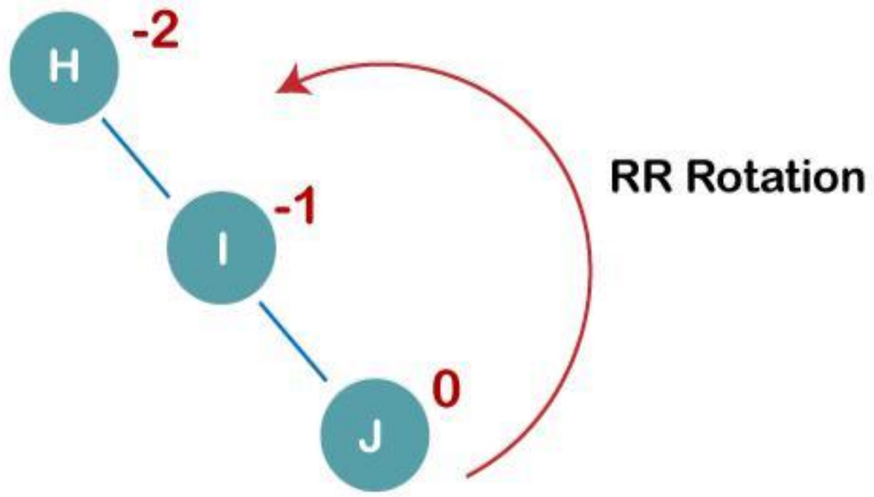




Character	Code
M	1010
A	0
H	110
R	111
S	1011
T	100

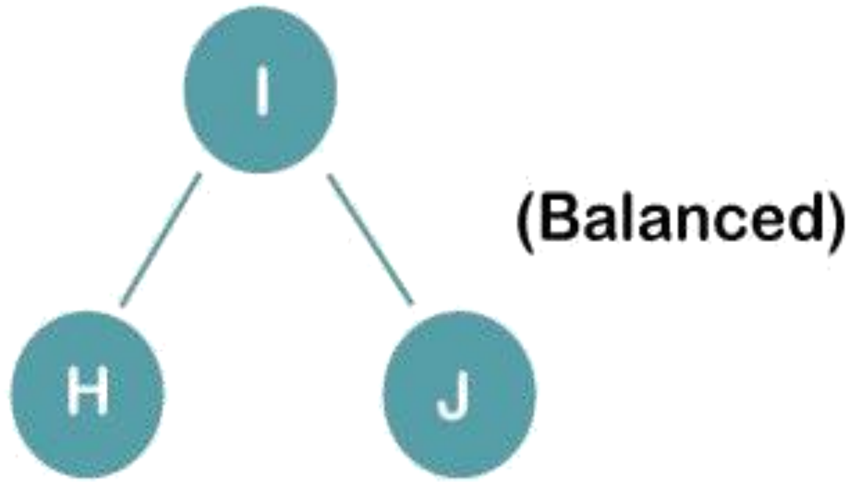
Qustion: Construct an AVL tree having the following elements H,I,J,B,A,E,C,F,D,G,K,L

Solution: Step-1. Insert H, I, J

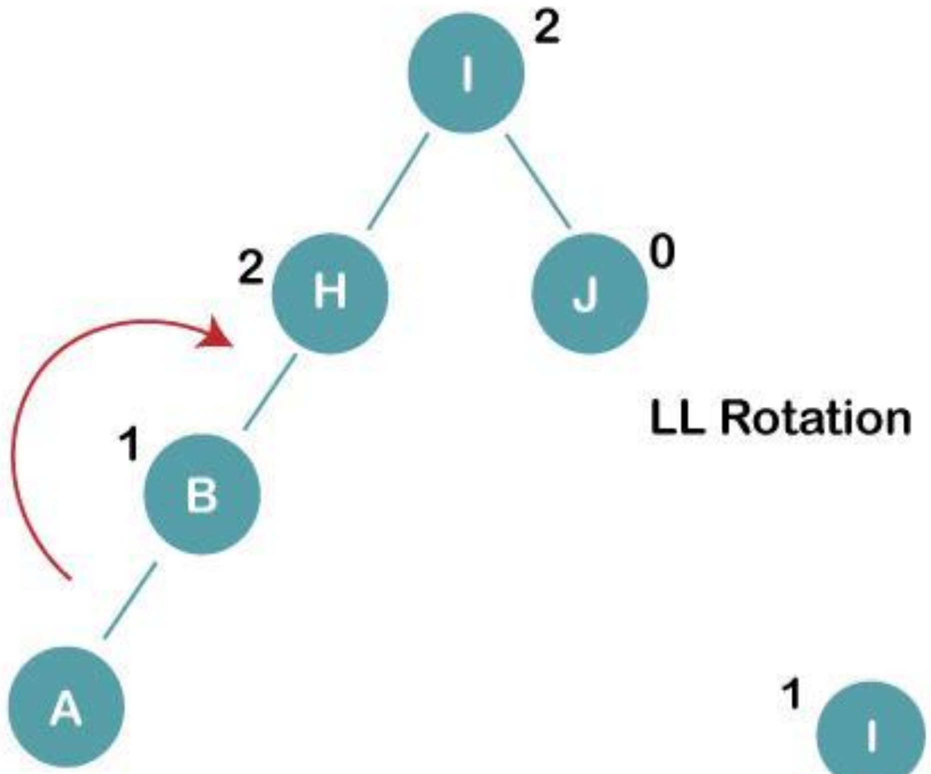


On inserting the above elements, especially in the case of H, the BST becomes unbalanced as the Balance Factor of H is -2. Since the BST is right-skewed, we will perform RR Rotation on node H.

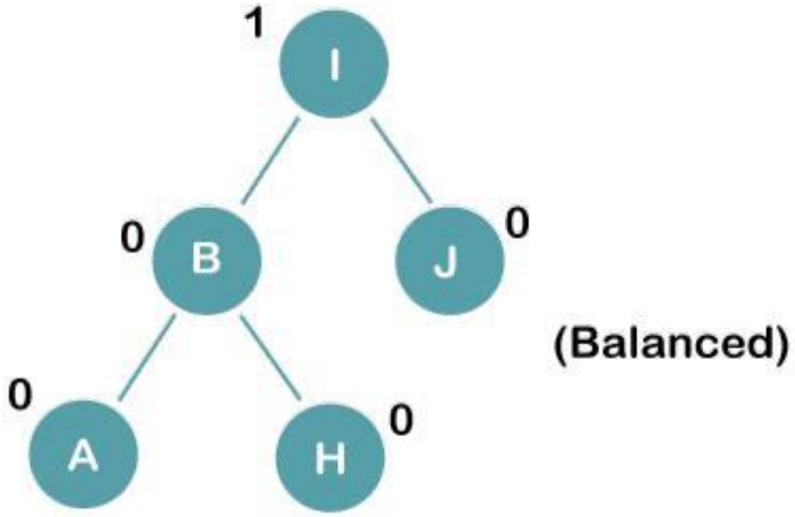
The resultant balance tree is:



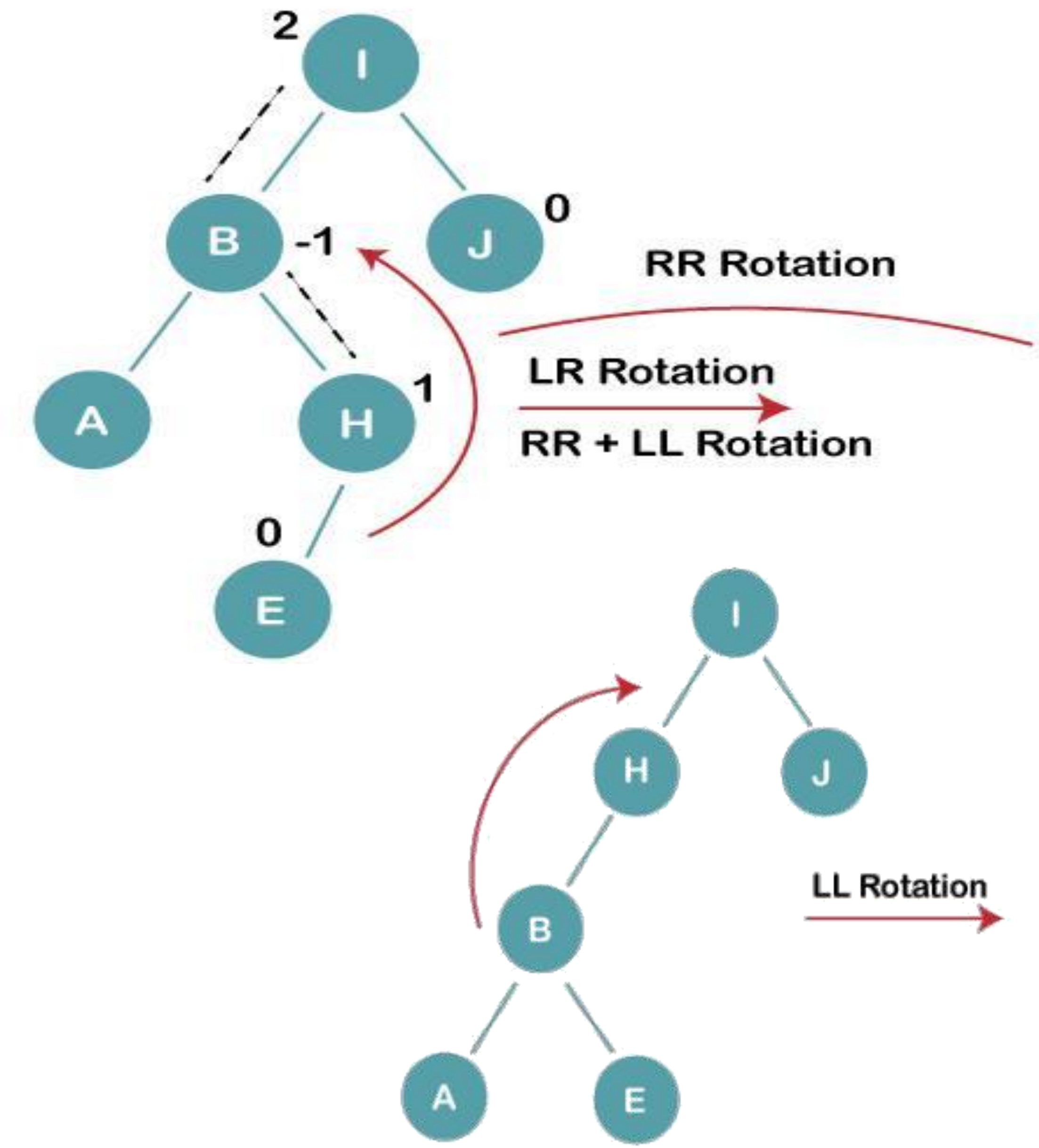
2. Insert B, A



On inserting the above elements, especially in case of A, the BST becomes unbalanced as the Balance Factor of H and I is 2, we consider the first node from the last inserted node i.e. H. Since the BST from H is left-skewed, we will perform LL Rotation on node H. The resultant balance tree is:



3. Insert E

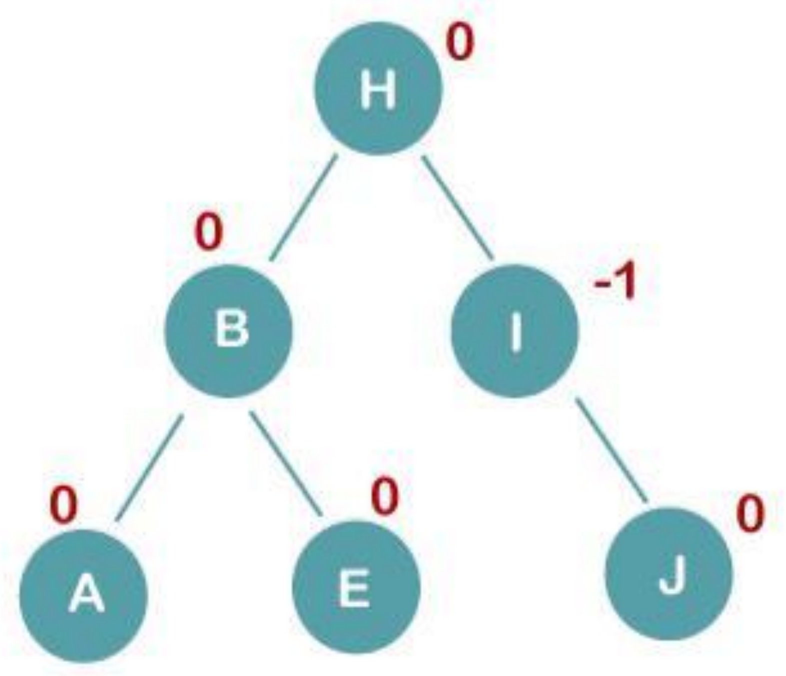


On inserting E, BST becomes unbalanced as the Balance Factor of I is 2, since if we travel from E to I we find that it is inserted in the left subtree of right subtree of I, we will perform LR Rotation on node I. LR = RR + LL rotation

3 a) We first perform RR rotation on node B

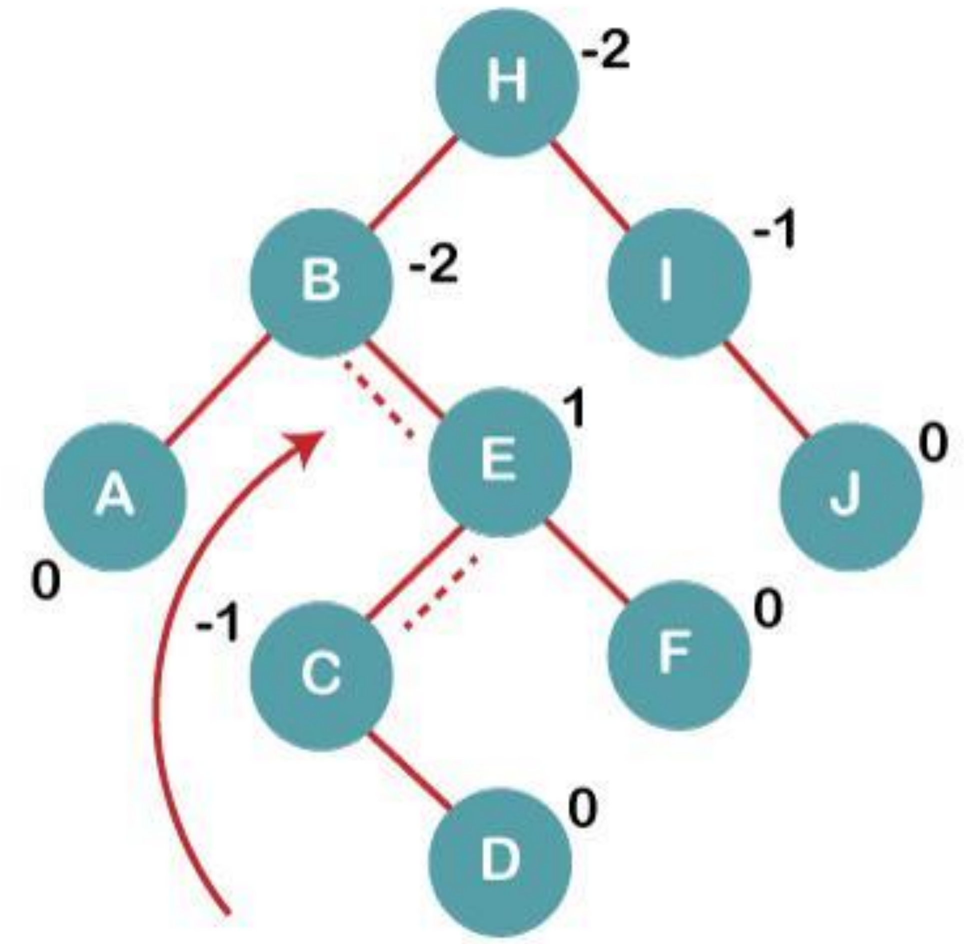
The resultant tree after RR rotation is:

3b) We first perform LL rotation on the node I, The resultant balanced tree after LL rotation is:



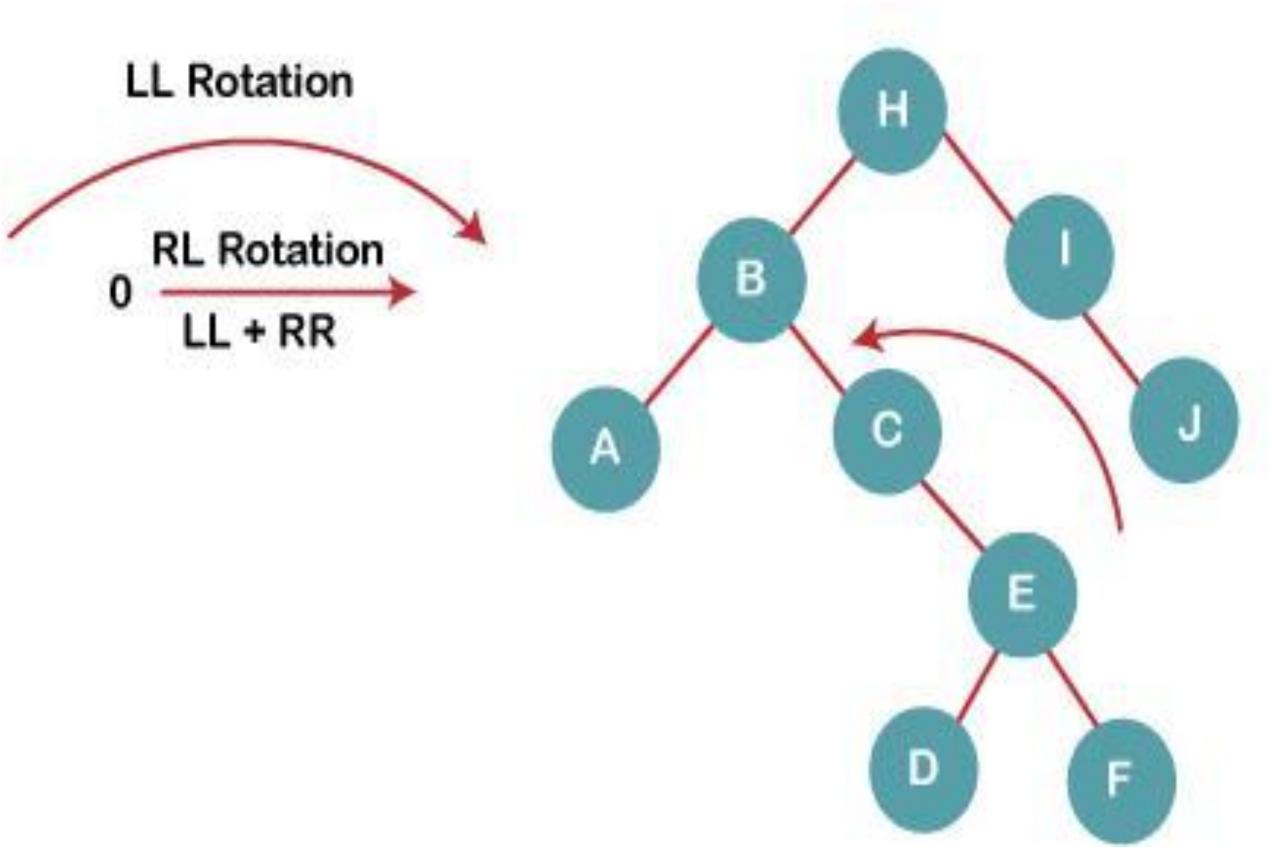
(Balanced)

4. Insert C, F, D



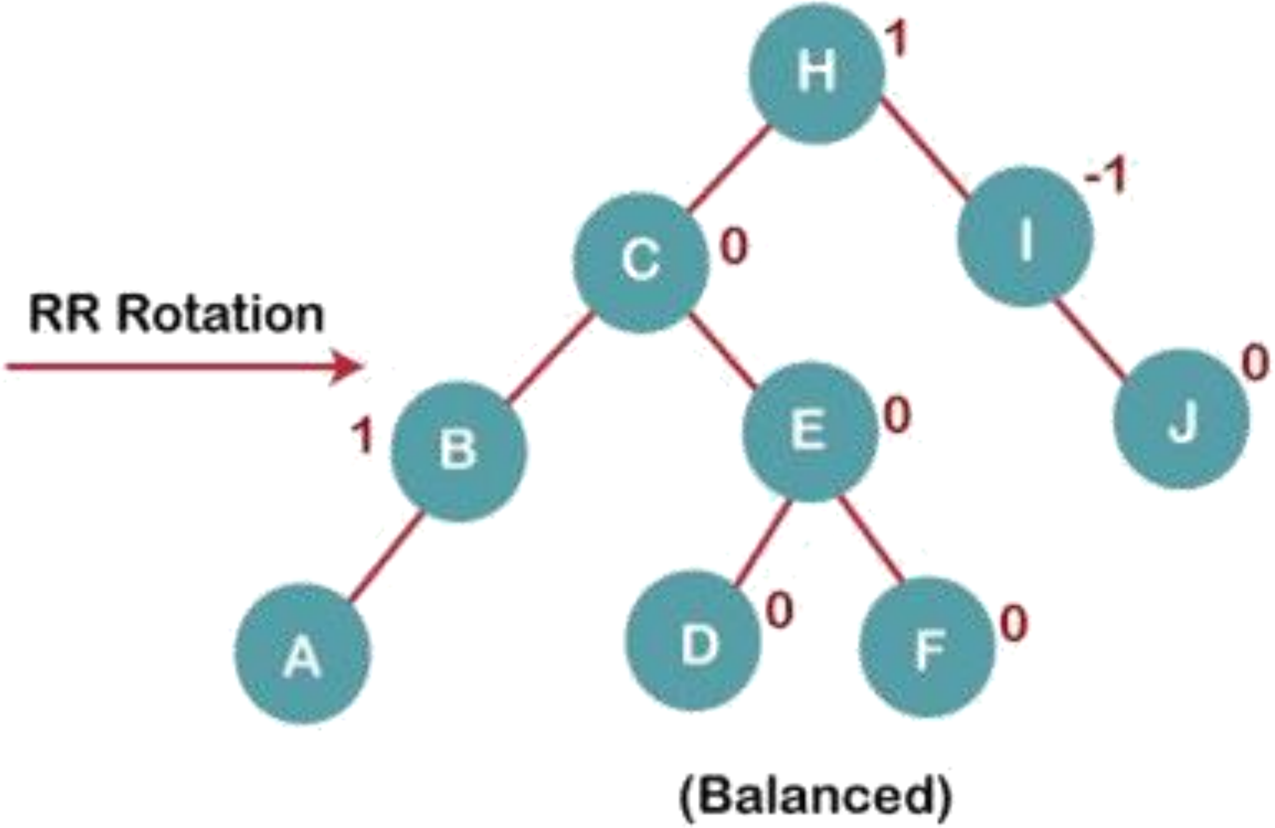
On inserting C, F, D, BST becomes unbalanced as the Balance Factor of B and H is -2, since if we travel from D to B we find that it is inserted in the right subtree of left subtree of B, we will perform RL Rotation on node I. RL = LL + RR rotation.

4a) We first perform LL rotation on node E, The resultant tree after LL rotation is:

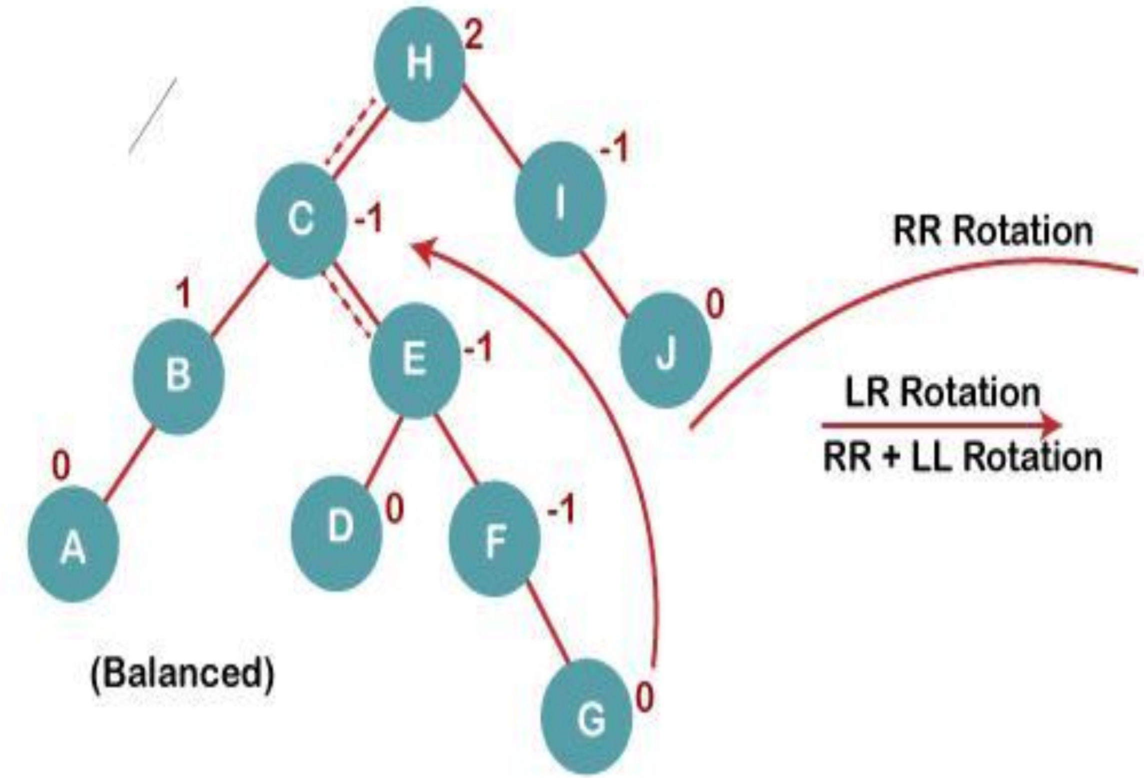


4b) We then perform RR rotation on node B

The resultant balanced tree after RR rotation is:



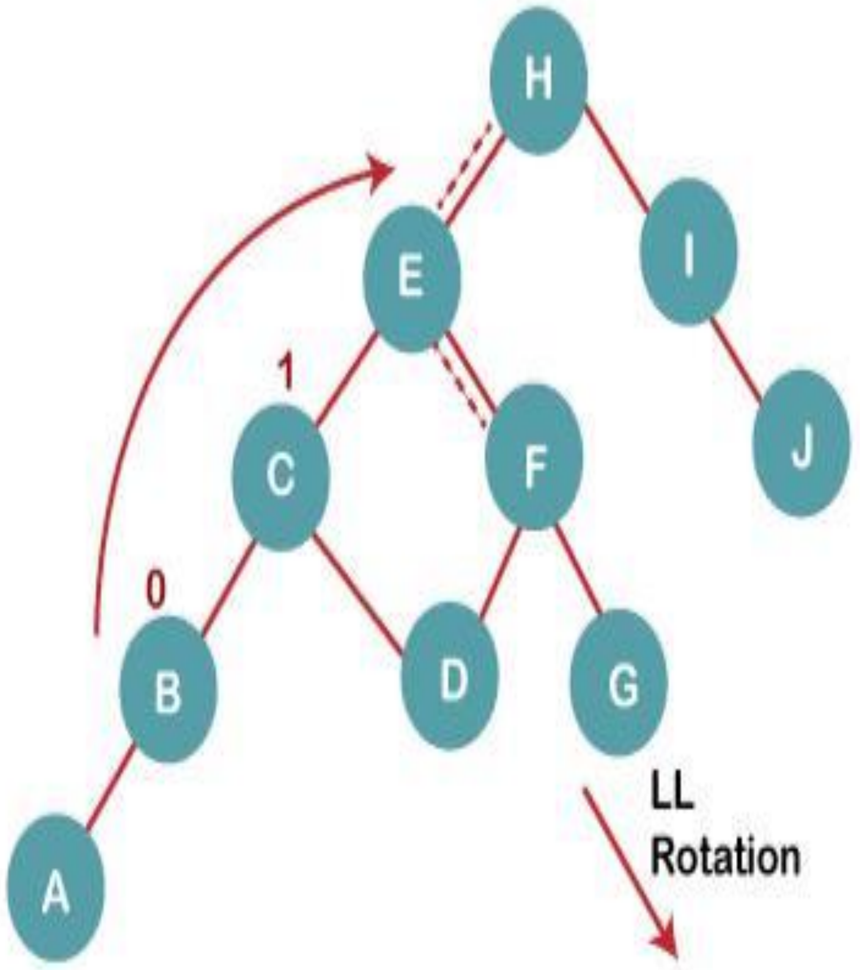
5. Insert G



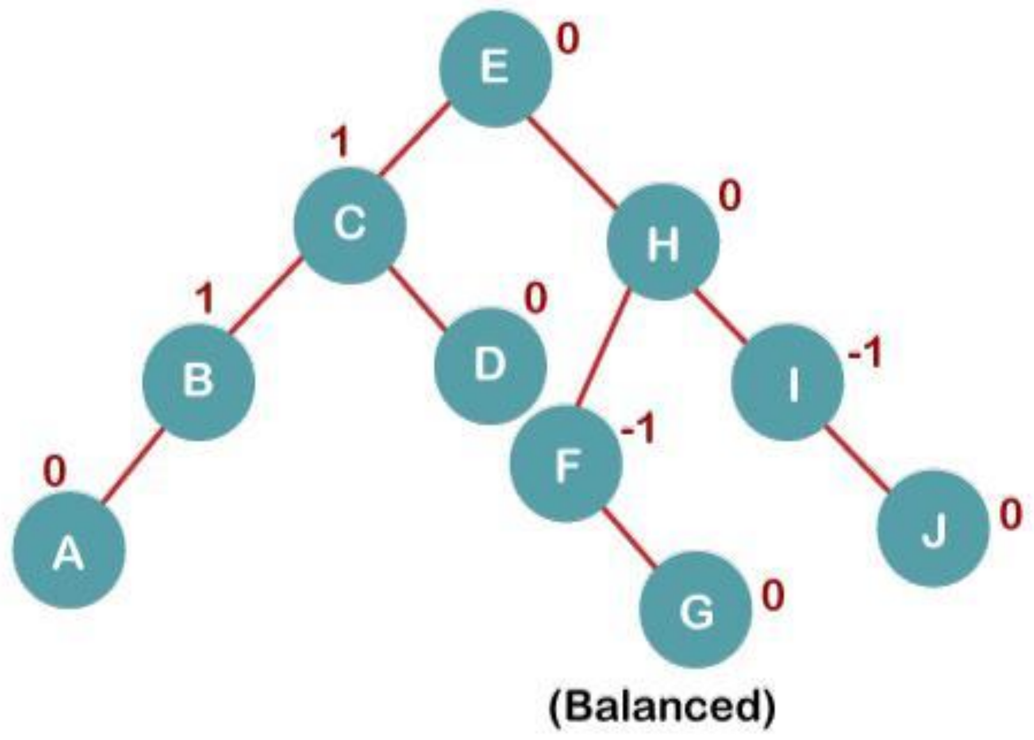
On inserting G, BST become unbalanced as the Balance Factor of H is 2, since if we travel from G to H, we find that it is inserted in the left subtree of right subtree of H, we will perform LR Rotation on node I. LR = RR + LL rotation.

5 a) We first perform RR rotation on node C

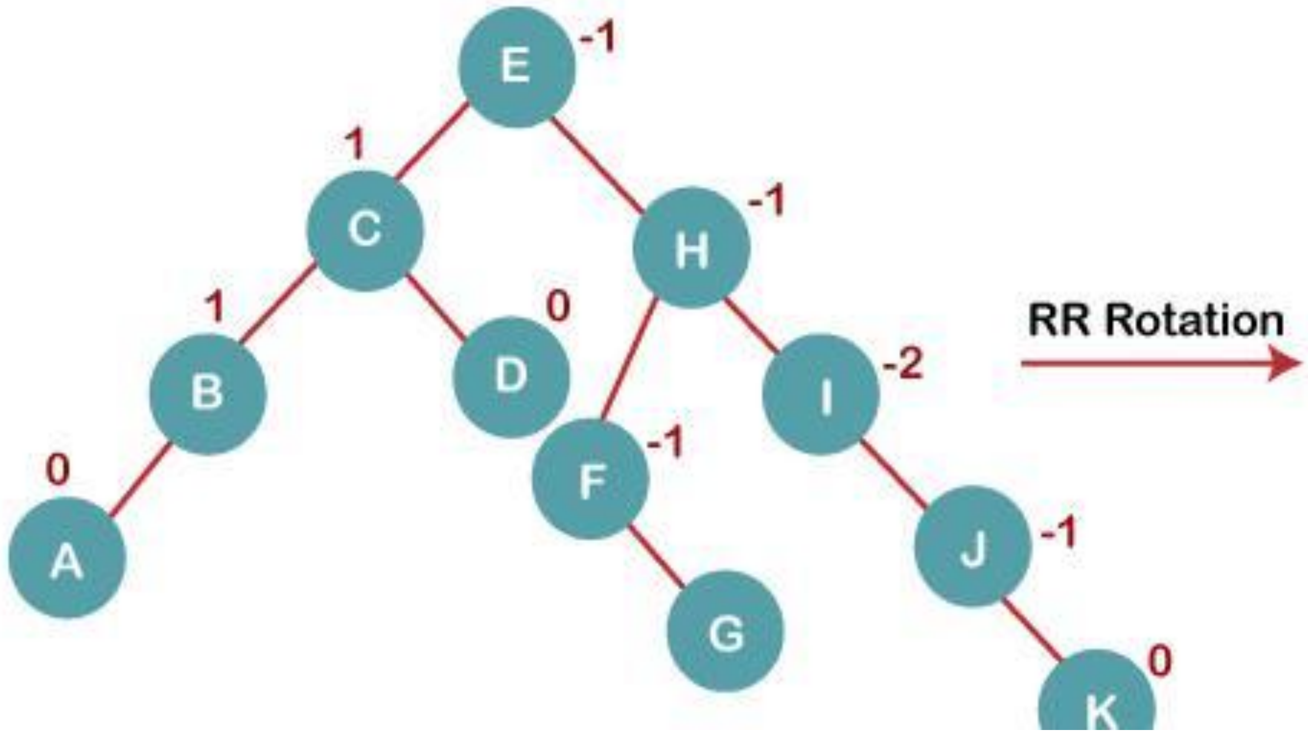
The resultant tree after RR rotation is:



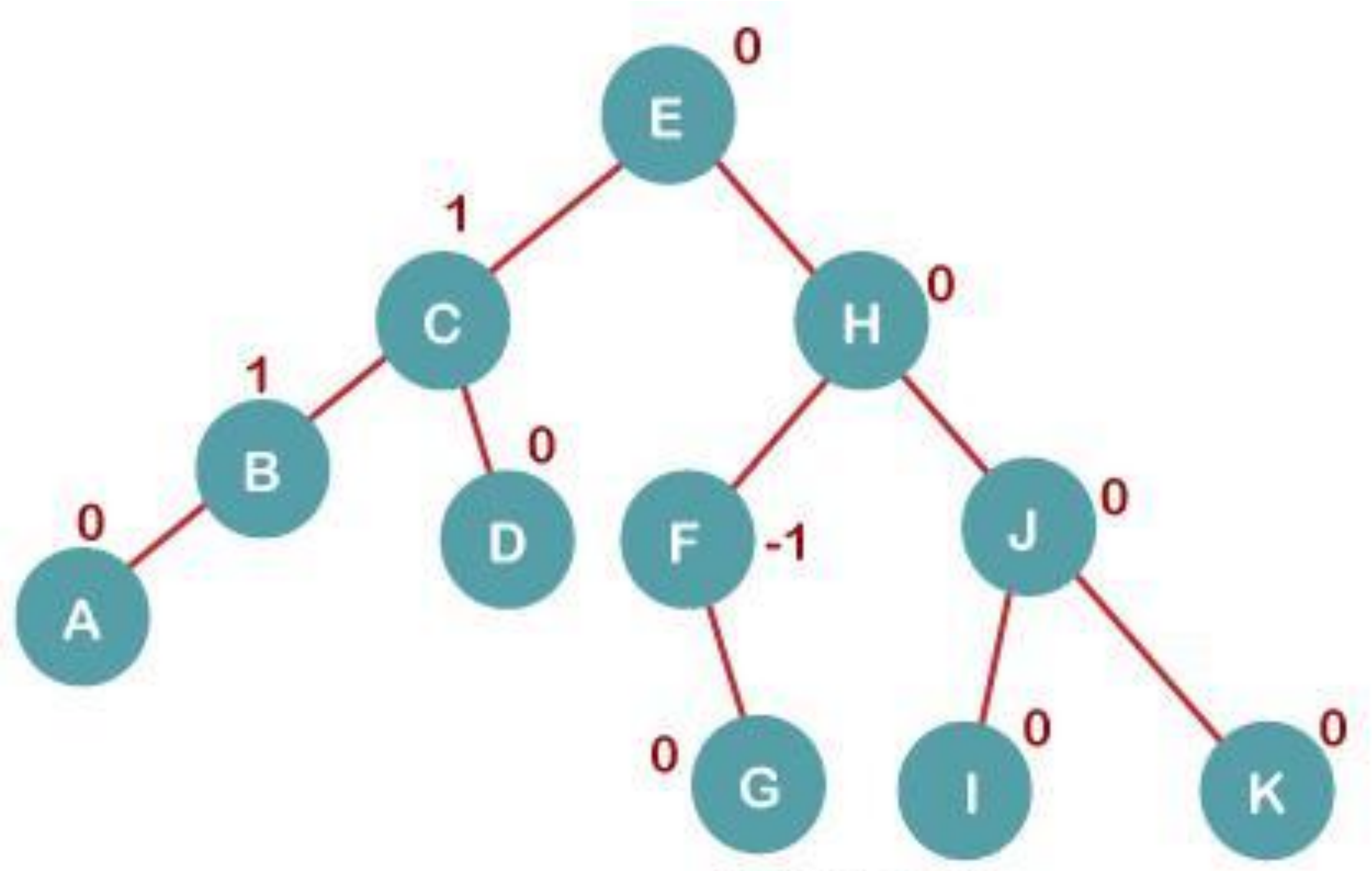
5 b) We then perform LL rotation on node H, The resultant balanced tree after LL rotation is:



6. Insert K



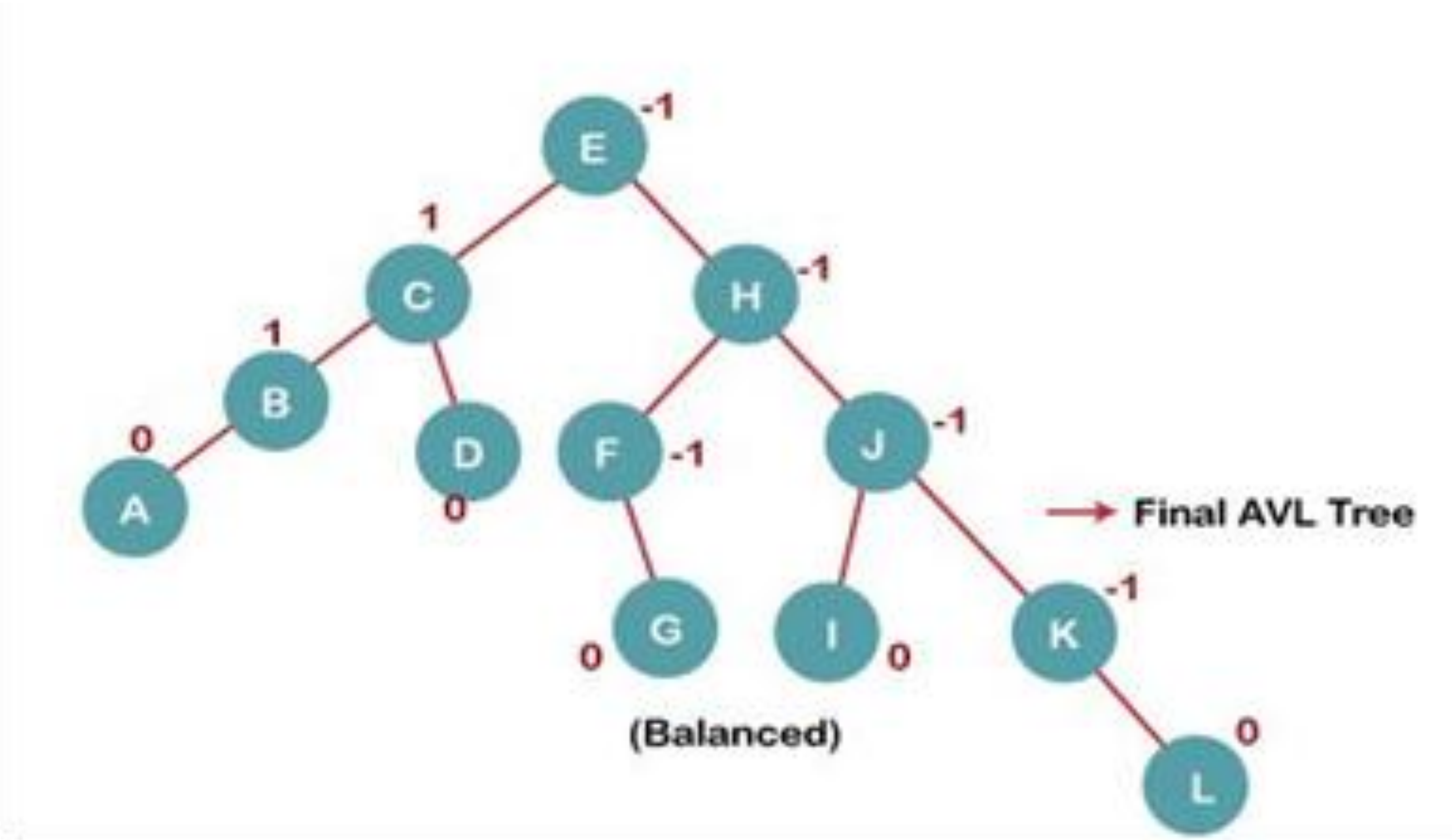
On inserting K, BST becomes unbalanced as the Balance Factor of I is -2. Since the BST is right-skewed from I to K, hence we will perform RR Rotation on the node I. The resultant balanced tree after RR rotation is:



(Balanced)

7. Insert L

On inserting the L tree is still balanced as the Balance Factor of each node is now either, -1, 0, +1. Hence the tree is a Balanced AVL tree

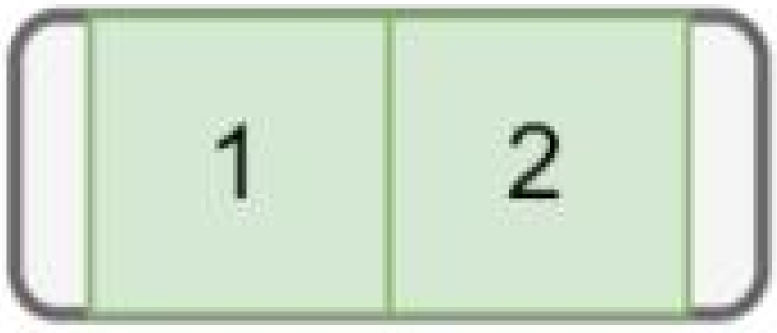


Question: Insert 10 nodes in a empty B-tree: (1,2,3,4,5,6,7,8,9,10), with the order of B-tree is 3.

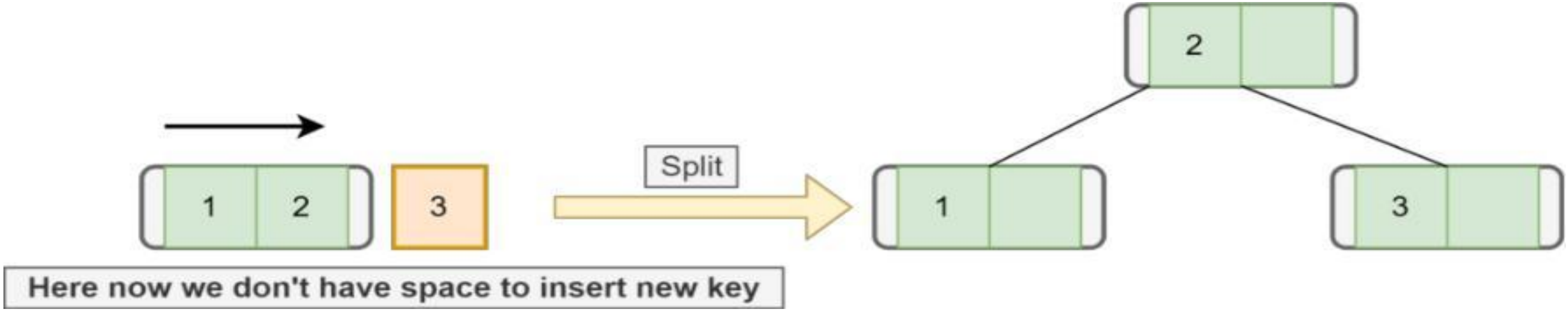
Solution: The maximum number of children a node can contain is 3. Additionally, the maximum and the minimum number of keys for a node, in this case, would be 2 and 1 (rounded off).

While inserting any element, we must remember that we can only insert values in ascending order. Moreover, we always insert elements at the leaf node so that the B-tree always grows in the upward direction.

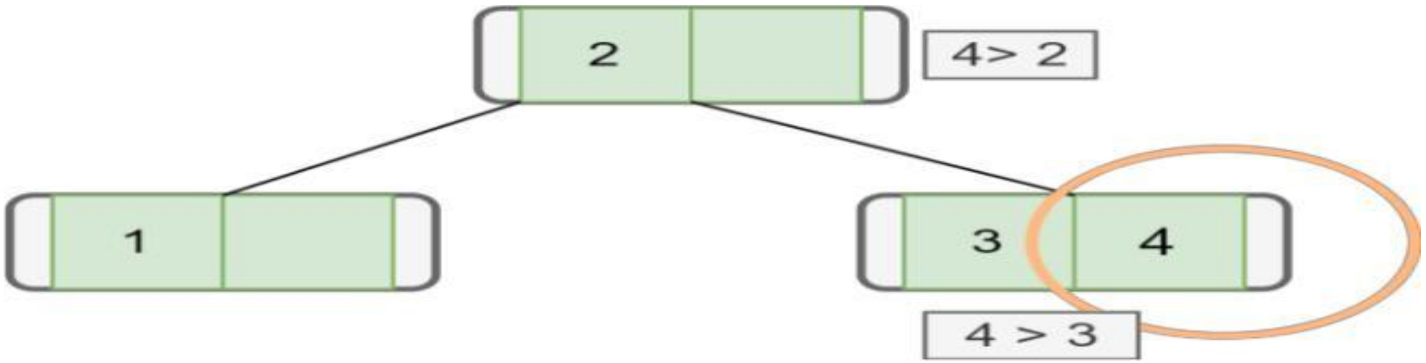
Let's start the insertion process with nodes 1 and 2 **:**



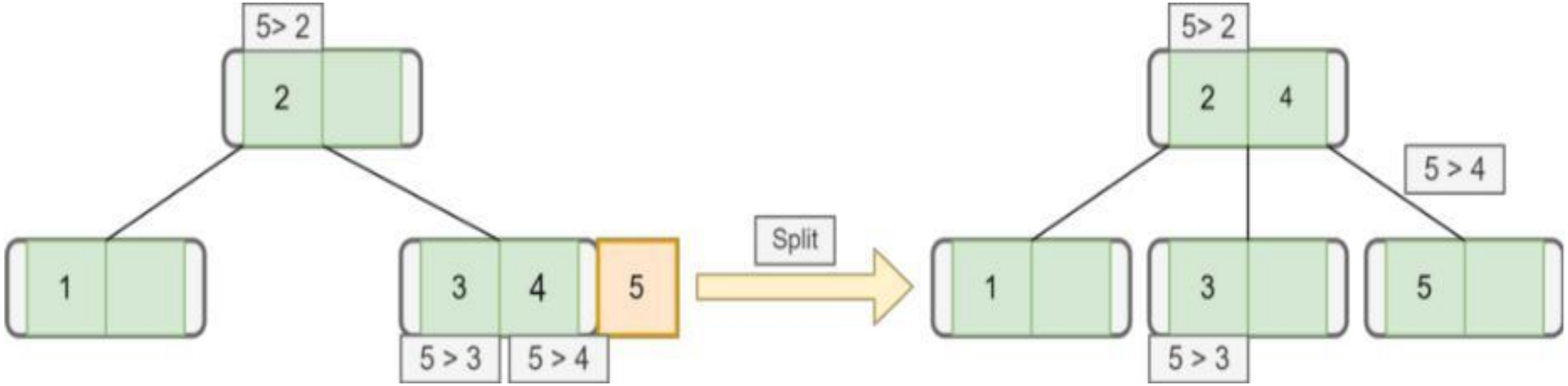
Next, **we want to inset node** 3. But there’s no space for inserting new elements in that leaf node as the maximum key a node can contain is here. Therefore, we need to split the node. To split the node, we take the median key and move that key upward:



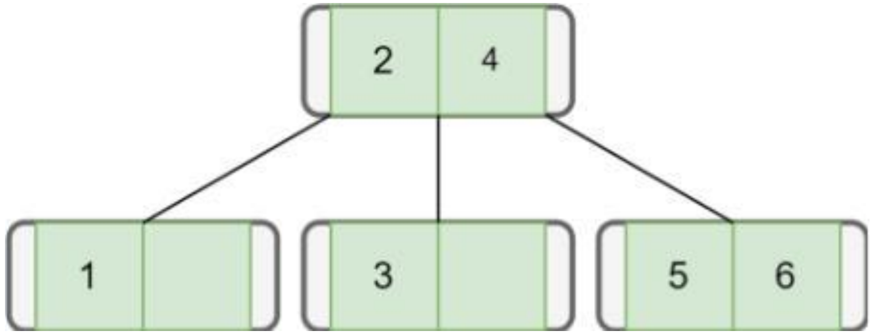
Now **we want to insert the next element, which is** subtree. In the right subtree, there’s a node with key Hence, we’ll add this node to the right of :



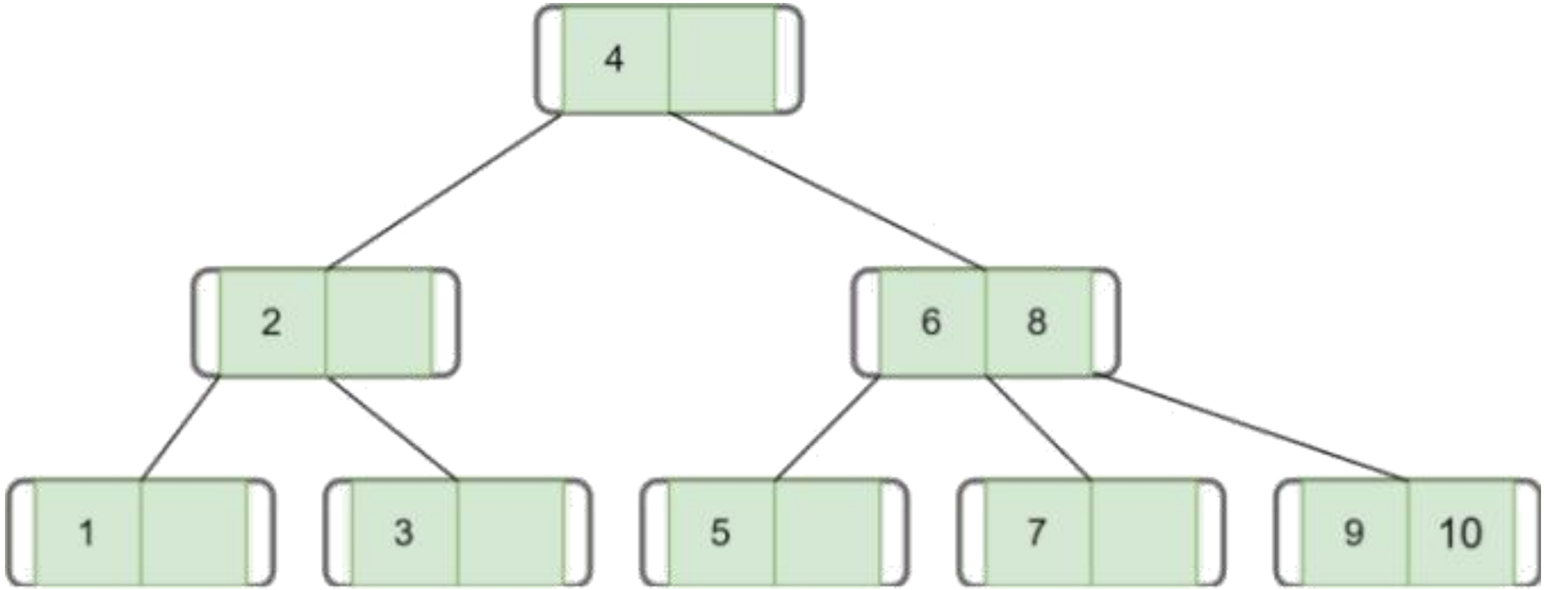
Let’s insert the next node with key 5. We first check the root, and as is greater than , we go to the right subtree. In the right subtree, we see is greater than and . Hence, we insert on the right side of . But again there’s not enough space. Therefore, we split it and move the median of that is, to the upper node:



Similarly, **we insert our next node with key** 6:



Finally, let's add all the remaining nodes one by one following the properties of the B-tree:

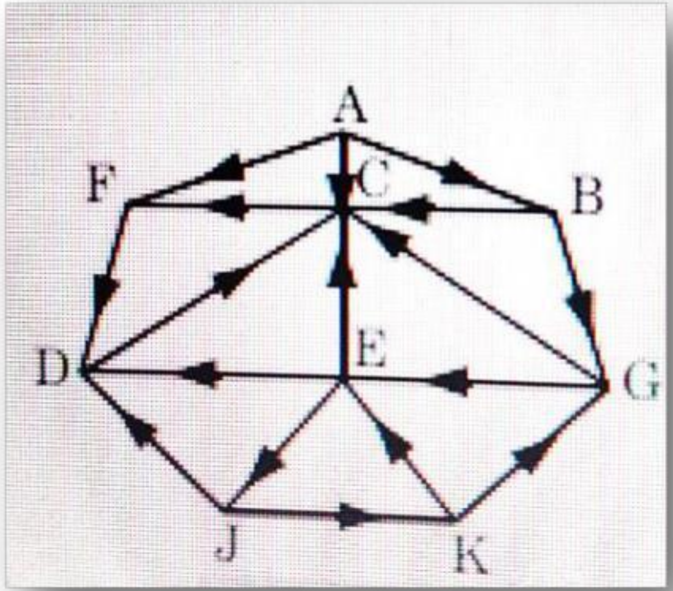


Solved Numerical
UNIT-5

Question : Implement BFS algorithm to find the shortest path from node *A* to *J*.

Solution: **Adjacency list of the graph is :**

A	:F, C, B
B	:G, C
C	:F
D	:C
E	:D, C, J
F	:D
G	:C, E
J	:D, K
K	:E, G

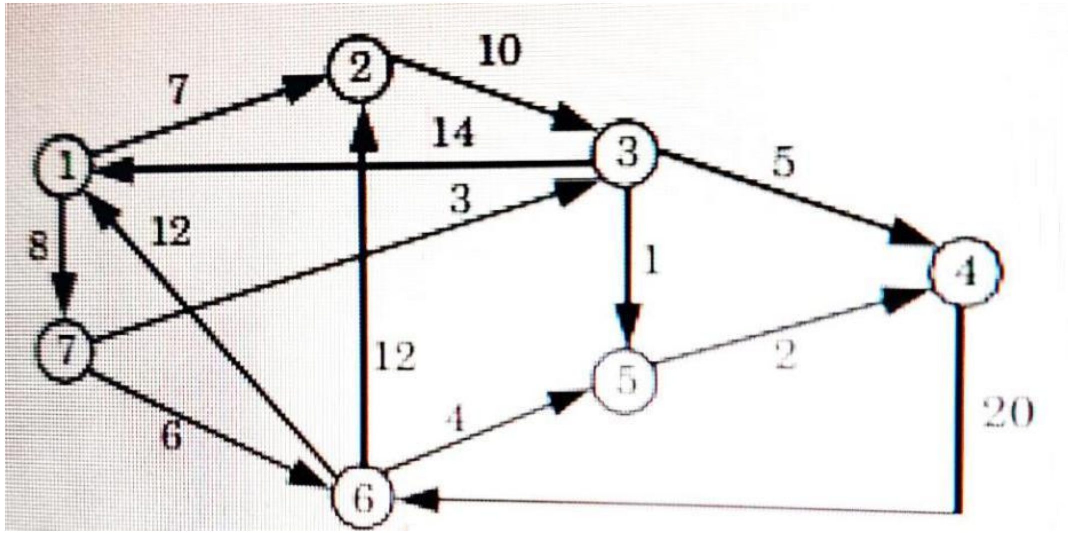


J is our final destination. We now back track from J to find the path from J to A : J← E ←G← B ←A

1. Initially set STATUS=1 for all vertex.
2. Now add ‘A’ to Queue and set STATUS = 2 Queue: A
3. Remove A from Queue and set STATUS=3. And add F,C,B in Queue and change STATUS=2
4. Remove F from Queue , add D in Queue, Now BFS = A, Queue = C, B, D,
5. Remove C from Queue, add F, but F is already visited. So no vertex will be added in this step.
   BFS = A, F, Queue = B, D
6. Remove B from Queue, add G, BFS = A, F, C, B, Queue = D, G
7. Remove D from Queue, BFS = A, F, C, B, D, Queue = G
8. Remove G from Queue, add E, BFS = A, F, C, B, D, G, Queue = E
9. Remove E from Queue, add J, BFS = A, F, C, B, D, G, E, Queue = J
10. Remove J, BFS=A,F,C,B,D,G,E,J Queue = Empty

J is our final destination. We now back track from J to find the path from J to A : J← E ←G← B ←A

Question : Implement DFS algorithm in the graph.

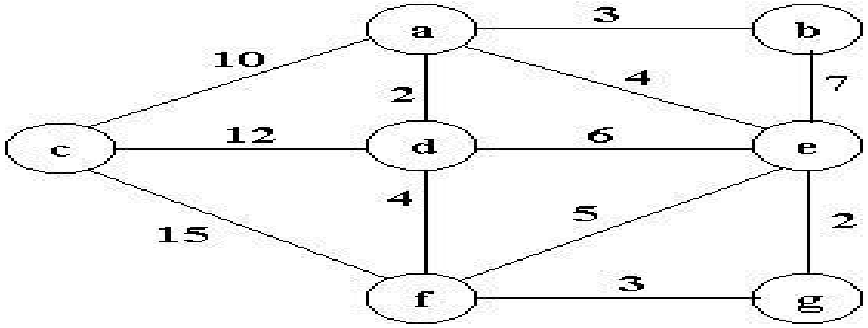


Numerical : Adjacency list of the given graph :

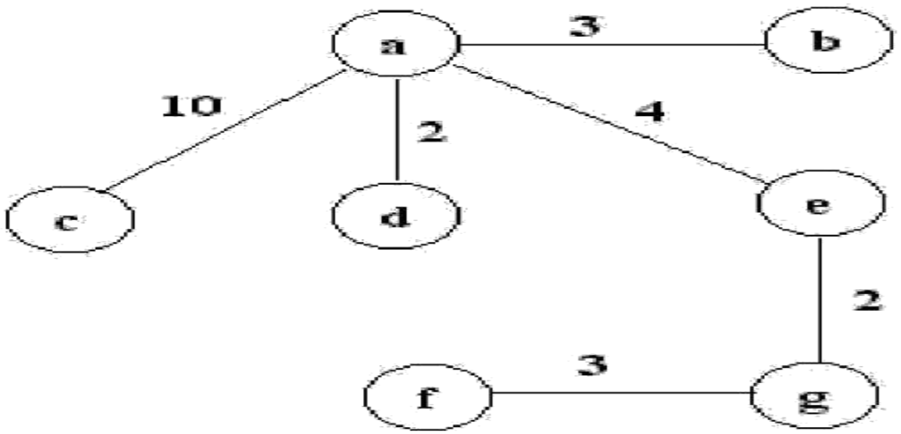
1→ 2,7
2→ 3
3→ 5,4,1
4→ 6
5→ 4
6→ 2,5,1
7→ 3,6

1. Initially set STATUS = 1 for all vertex, Push 1 onto stack and set their STATUS = 2, Stack: 1
2. Pop 1 from stack, change its STATUS = 1 and Push 2, 7 onto stack and change their STATUS = 2;
DFS =1 Stack: 7
2
3. Pop 7 from stack, Push 3, 6; DFS = 1, 7 Stack: 6
3
2
4 . Pop 6 from stack, Push 5; DFS=1,7,6 Stack: 5
3
2
5. Pop 5 from stack, Push 4; DFS=1,7,6,5 Stack: 4
3
2
6. Pop 4 from stack; DFS=1,7,6,5,4 Stack: 3
2
7. Pop 3 from stack; DFS=1,7,6,5,4,3 Stack: 2
8. Pop 2 from stack; DFS=1,7,6,5,4,3
Now, the stack is empty, so the depth first traversal of a given graph is 1, 7, 6, 5, 4, 3.

Question: Trace Kruskal's algorithm in finding a minimum-cost spanning tree for the undirected, weighted graph given below.

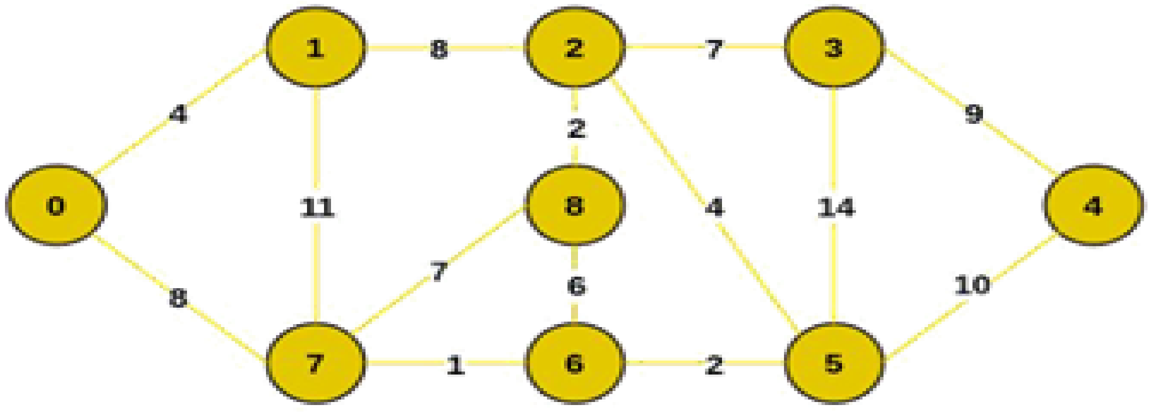


edge	ad	eg	ab	fg	ae	df	ef	de	be	ac	cd	cf
weight	2	2	3	3	4	4	5	6	7	10	12	15
insertion status	√	√	√	√	√	x	x	x	x	√	x	x
insertion order	1	2	3	4	5					6		



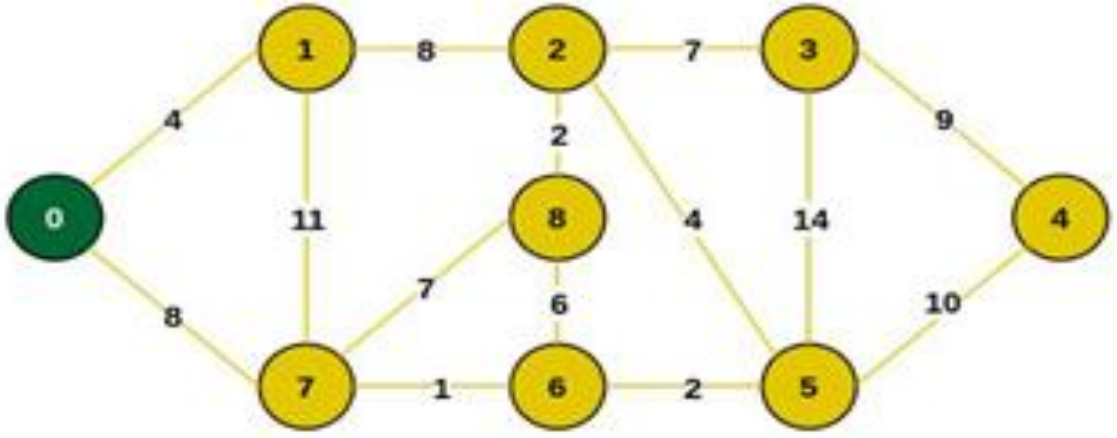
Therefore The minimum cost is: 24

Question: Consider the following graph as an example for which we need to find the Minimum Spanning Tree (MST).



Example of a Graph

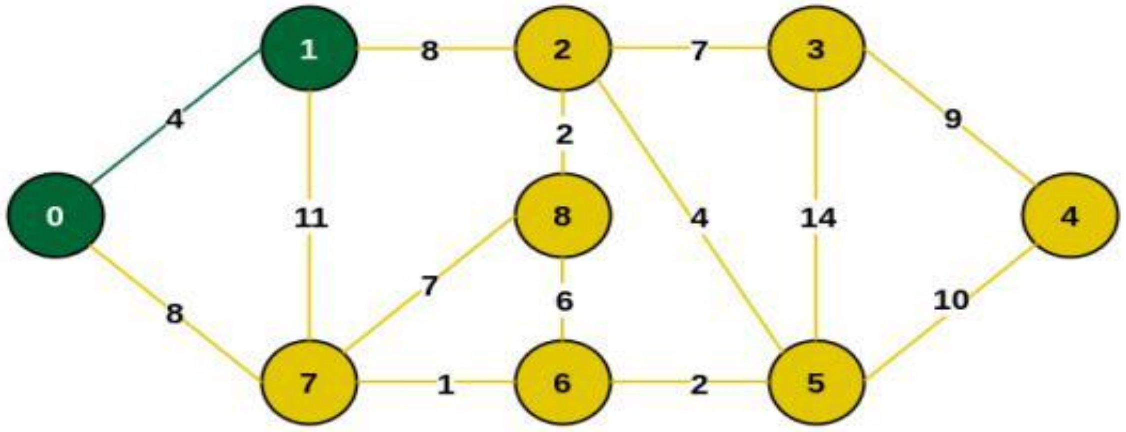
Step 1: Firstly, we select an arbitrary vertex that acts as the starting vertex of the Minimum Spanning Tree. Here we have selected vertex 0 as the starting vertex.



Select an arbitrary starting vertex. Here we have selected 0

0 is selected as starting vertex

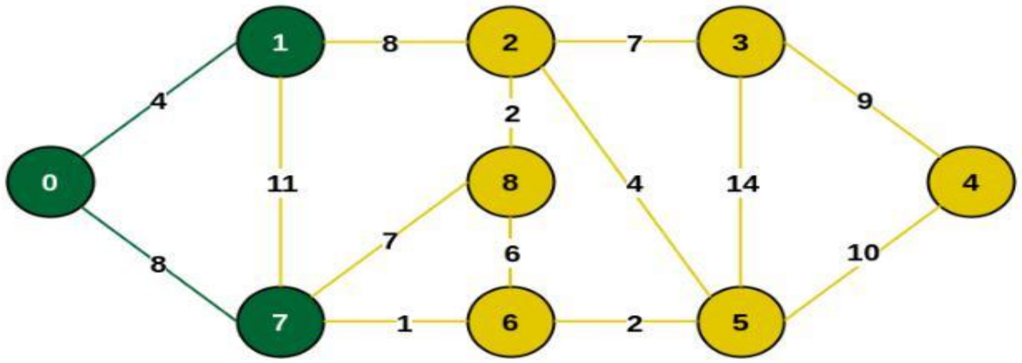
Step 2: All the edges connecting the incomplete MST and other vertices are the edges {0, 1} and {0, 7}. Between these two the edge with minimum weight is {0, 1}. S include the edge and vertex 1 in the MST.



Minimum weighted edge from MST to other vertices is 0-1 with weight 4

1 is added to the MST

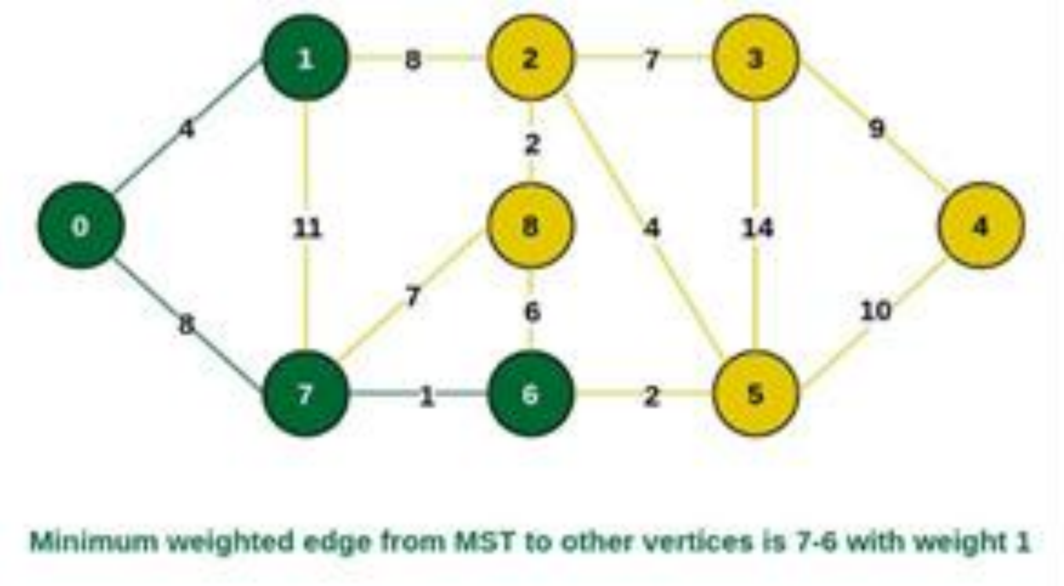
Step 3: The edges connecting the incomplete MST to other vertices are {0, 7}, {1, 7} and {1, 2}. Among these edges the minimum weight is 8 which is of the edges {0, 7} and {1, 2}. Let us here include the edge {0, 7} and the vertex 7 in the MST. [We could have also included edge {1, 2} and vertex 2 in the MST].



Minimum weighted edge from MST to other vertices is 0-7 with weight 8

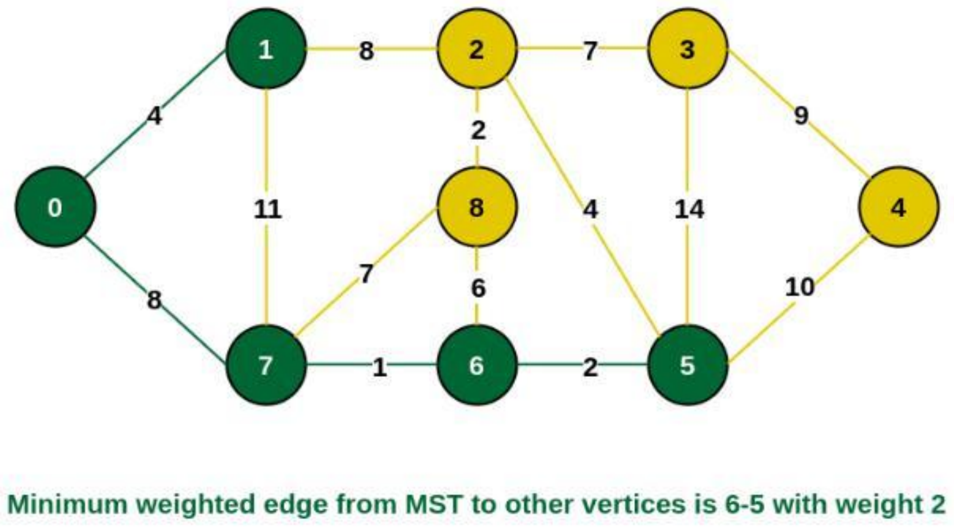
7 is added in the MST

Step 4: The edges that connect the incomplete MST with the fringe vertices are {1, 2}, {7, 6} and {7, 8}. Add the edge {7, 6} and the vertex 6 in the MST as it has the least weight (i.e., 1).



6 is added in the MST

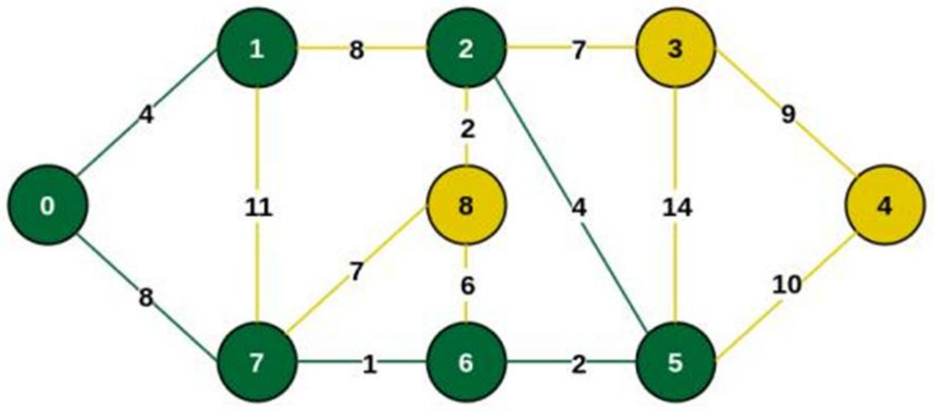
Step 5: The connecting edges now are {7, 8}, {1, 2}, {6, 8} and {6, 5}. Include edge {6, 5} and vertex 5 in the MST as the edge has the minimum weight (i.e., 2) among them.



Include vertex 5 in the MST

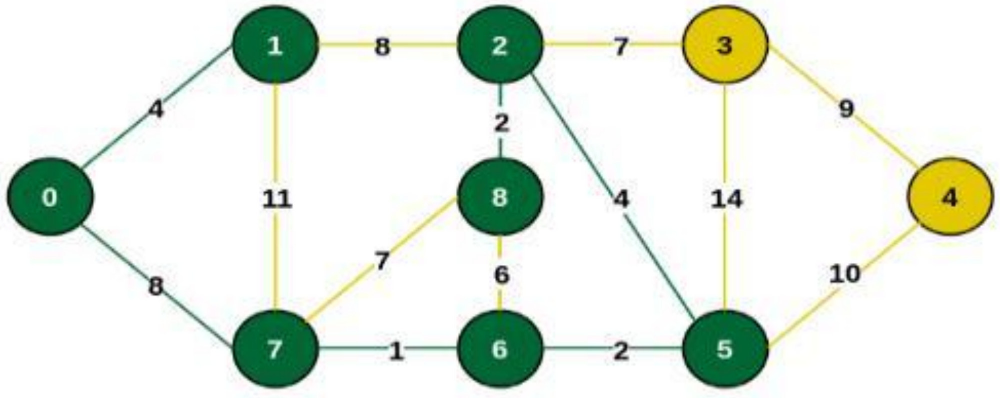
Step 6: Among the current connecting edges, the edge {5, 2} has the minimum weight. So include that edge and the vertex 2 in the MST.

Step 7: The connecting edges between the incomplete MST and the other edges are {2, 8}, {2, 3}, {5, 3} and {5, 4}. The edge with minimum weight is edge {2, 8} which has weight 2. So include this edge and the vertex 8 in the MST.



Minimum weighted edge from MST to other vertices is 5-2 with weight 4

Include vertex 2 in the MST

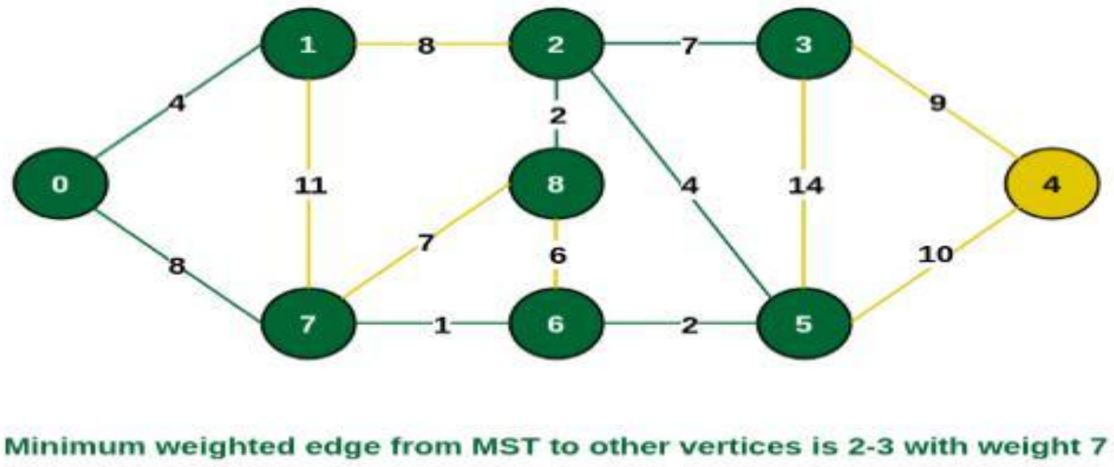


Minimum weighted edge from MST to other vertices is 2-8 with weight 2

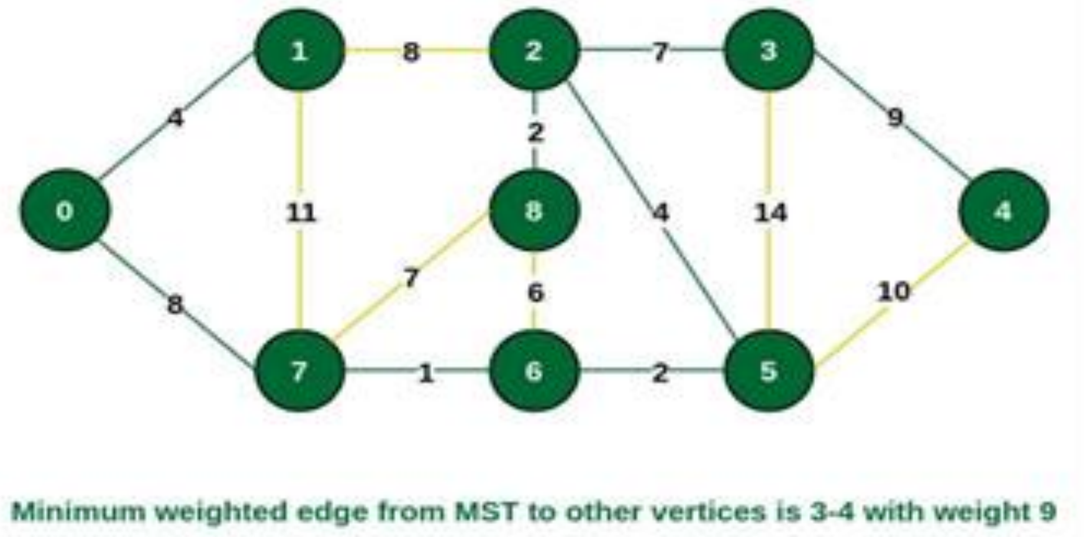
Add vertex 8 in the MST

Step 8: See here that the edges {7, 8} and {2, 3} both have same weight which are minimum. But 7 is already part of MST. So we will consider the edge {2, 3} and include that edge and vertex 3 in the MST.

Step 9: Only the vertex 4 remains to be included. The minimum weighted edge from the incomplete MST to 4 is {3, 4}.

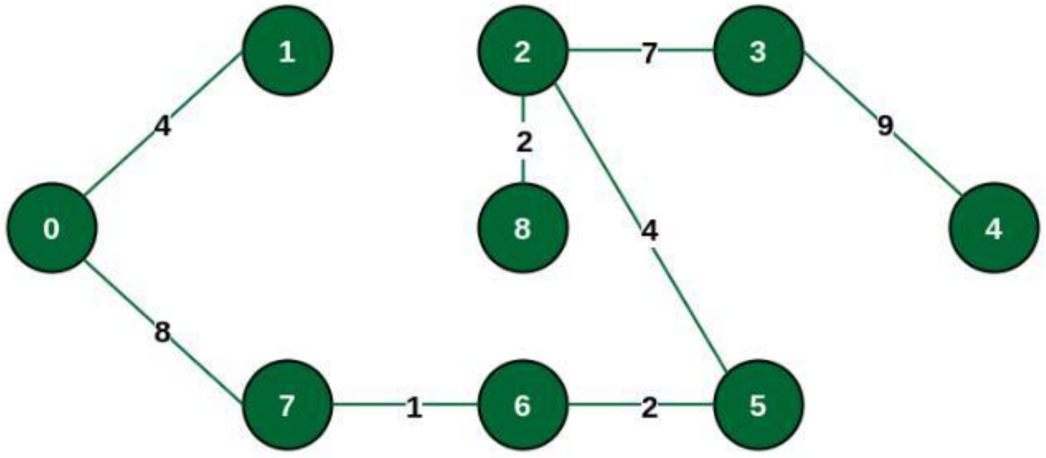


Include vertex 3 in MST



Include vertex 4 in the MST

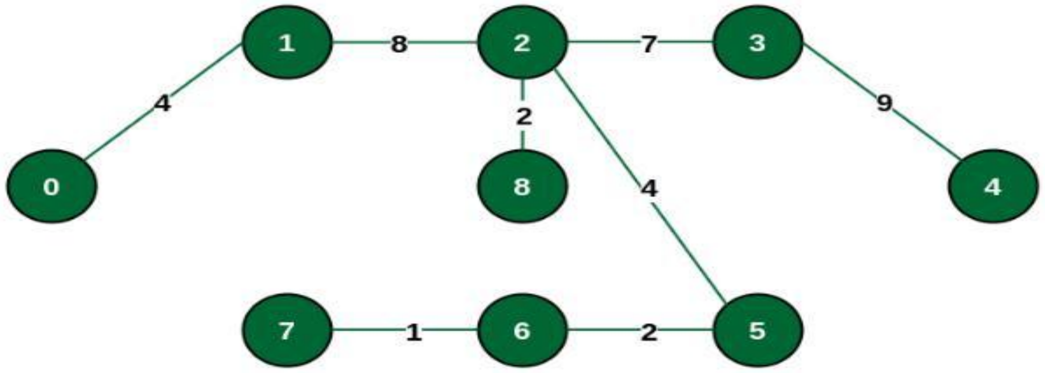
The final structure of the MST is as follows and the weight of the edges of the MST is (4 + 8 + 1 + 2 + 4 + 2 + 7+9) = **37**.



The final structure of MST

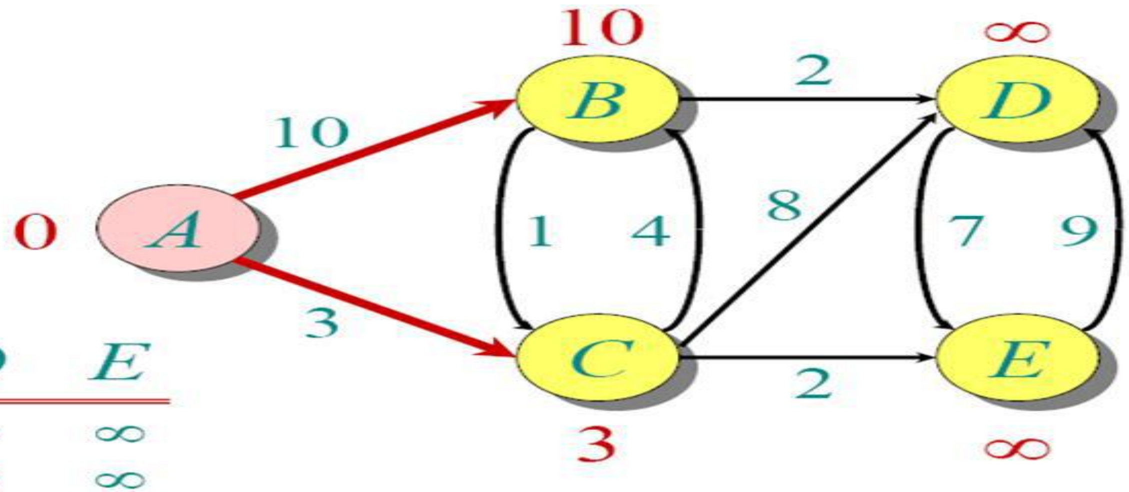
The structure of the MST formed using the above method

Note: If we had selected the edge {1, 2} in the third step then the MST would look like the following.



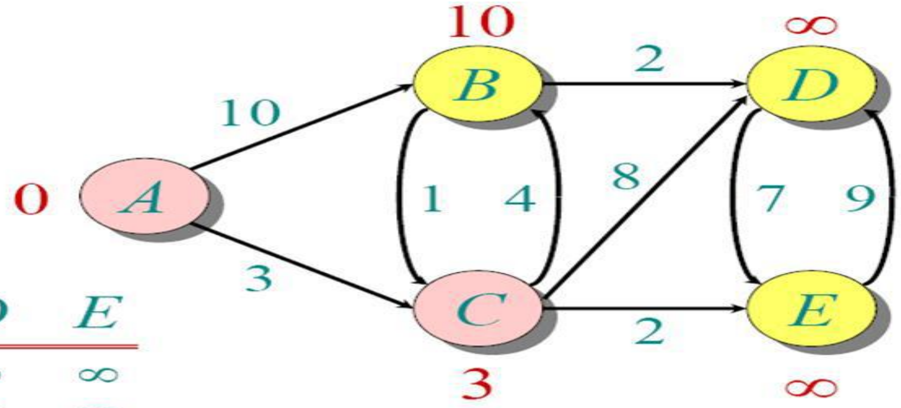
Alternative MST structure

Question:: Consider A as a source vertex. Find the Shortest path from source to all nodes.



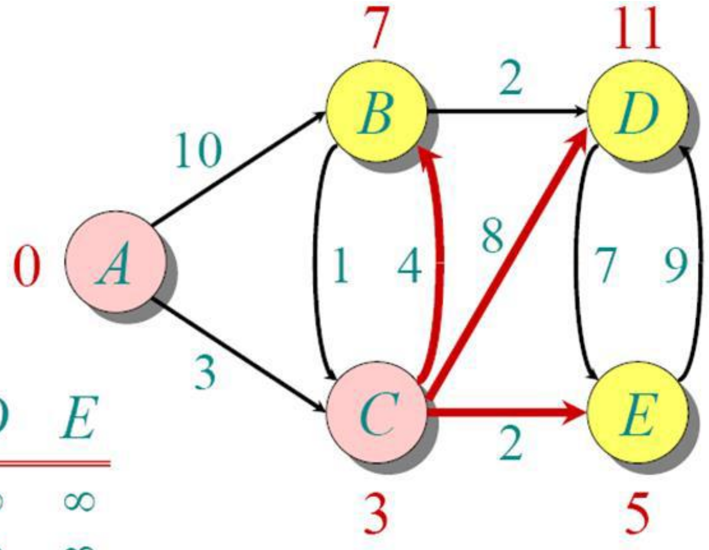
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞

S: { A }



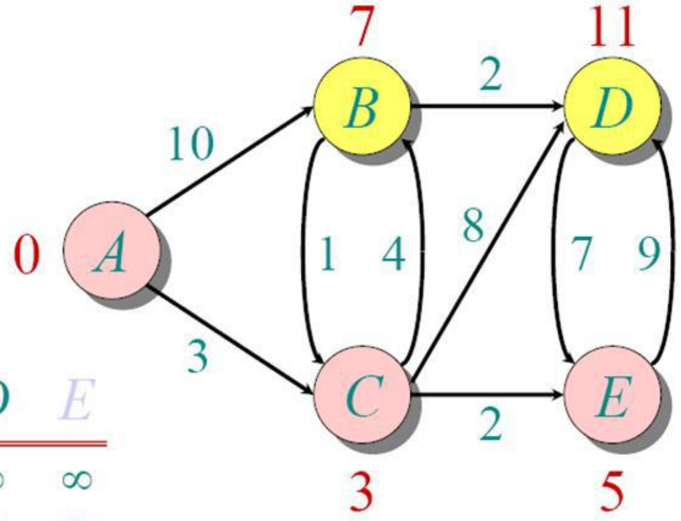
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞

S: { A, C }



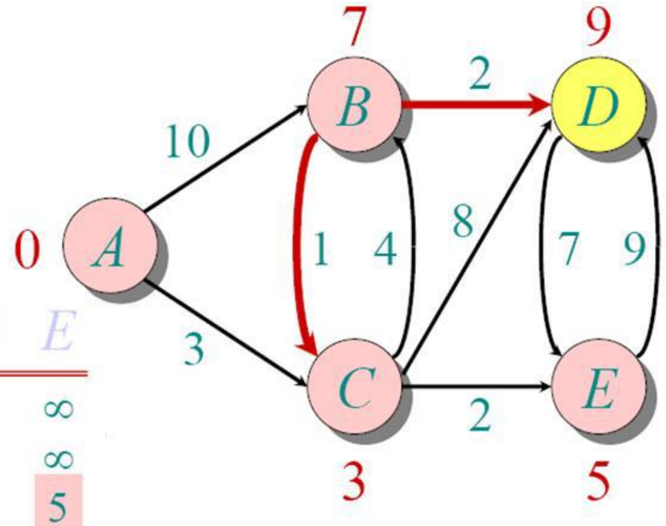
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5

S: { *A*, *C* }



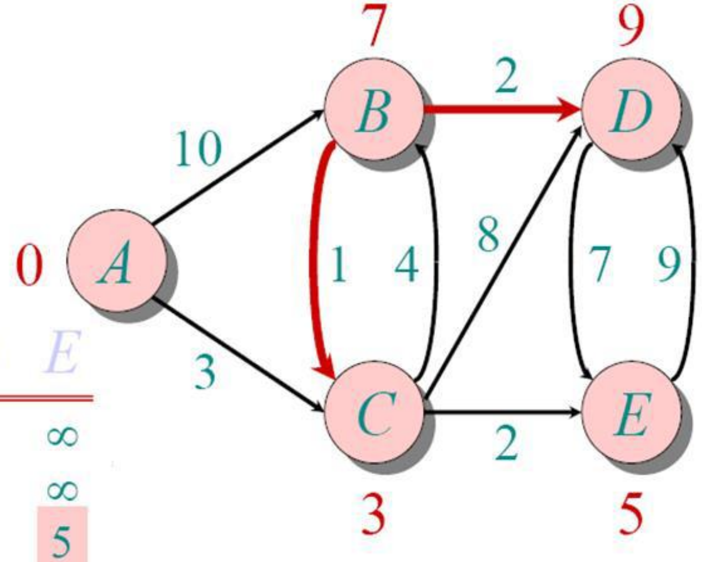
Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5

S: { *A*, *C*, *E* }



Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5
		7		11	
				9	

S: { A, C, E, B }



Q:	A	B	C	D	E
	0	∞	∞	∞	∞
		10	3	∞	∞
		7		11	5
		7		11	
				9	

S: { A, C, E, B, D }

Question: Consider the following directed weighted graph-Using Floyd Warshall Algorithm, find the shortest path distance between every pair of vertices.

Solution- Step-01:

Write the initial distance matrix.

- It represents the distance between every pair of vertices in the form of given weights.
- For diagonal elements (representing self-loops), distance value = 0.
- For vertices having a direct edge between them, distance value = weight of that edge.
- For vertices having no direct edge between them, distance value = ∞.

Initial distance matrix for the given graph is-

$$D_0 = \begin{array}{c c} & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & 8 & \infty & 1 \\ \infty & 0 & 1 & \infty \\ 4 & \infty & 0 & \infty \\ \infty & 2 & 9 & 0 \end{bmatrix} \end{array}$$

Step-02:

Using Floyd Warshall Algorithm, write the following 4 matrices-

$$D_1 = \begin{array}{c c} & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & 8 & \infty & 1 \\ \infty & 0 & 1 & \infty \\ 4 & 12 & 0 & 5 \\ \infty & 2 & 9 & 0 \end{bmatrix} \end{array}$$

$$D_2 = \begin{array}{cc} & \begin{array}{cccc}1 & 2 & 3 & 4\end{array} \\ \begin{array}{c}1\\2\\3\\4\end{array} & \begin{bmatrix}0 & 8 & 9 & 1 \\ \infty & 0 & 1 & \infty \\ 4 & 12 & 0 & 5 \\ \infty & 2 & 3 & 0\end{bmatrix}\end{array}$$

$$D_3 = \begin{array}{cc} & \begin{array}{cccc}1 & 2 & 3 & 4\end{array} \\ \begin{array}{c}1\\2\\3\\4\end{array} & \begin{bmatrix}0 & 8 & 9 & 1 \\ 5 & 0 & 1 & 6 \\ 4 & 12 & 0 & 5 \\ 7 & 2 & 3 & 0\end{bmatrix}\end{array}$$

$$D_4 = \begin{array}{cc} & \begin{array}{cccc}1 & 2 & 3 & 4\end{array} \\ \begin{array}{c}1\\2\\3\\4\end{array} & \begin{bmatrix}0 & 3 & 4 & 1 \\ 5 & 0 & 1 & 6 \\ 4 & 7 & 0 & 5 \\ 7 & 2 & 3 & 0\end{bmatrix}\end{array}$$

Patterns of questions commonly asked in semester exams

Unit	No. of Questions From Section A (Each Ques. Carry 2 Marks)		No. of Questions From Section B (Each Ques. Carry 7 Marks)		No. of Questions From Section C (Each Ques. Carry 7 Marks)		Total Marks
	No of Ques.	Marks	No of Ques.	Marks	No of Ques.	Marks	
1	2	4	1	7	2	14	25
2	2	4	1	7	2	14	25
3	1	2	1	7	2	14	23
4	1	2	1	7	2	14	23
5	1	2	1	7	2	14	23

BTECH
(SEM III) THEORY EXAMINATION 2024-25
DATA STRUCTURE

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 07 = 14**

Q no.	Question	CO	Level
a.	Explain why do we need a pointer to maintain a linked list structure? समझाइये कि लिंक्ड सूची संरचना को बनाए रखने के लिए हमें पॉइंटर की आवश्यकता क्यों होती है?	CO1	K4
b.	If an array starts at address 2000 and each integer takes 4 bytes, evaluate the address of arr[5]? यदि एक सरणी पता 2000 से शुरू होती है और प्रत्येक पूर्णांक 4 बाइट्स लेता है, तो arr[5] का पता मूल्यांकन करें?	CO1	K5
c.	Explain Priority Queue and its significance? प्राथमिकता कतार और इसके महत्व को समझाइए?	CO2	K4
d.	The following sequence of operations is performed on stack: PUSH (10), PUSH (20), POP, PUSH (10), PUSH (20), POP, POP, POP, PUSH (20), POP. Demonstrate the sequence of the value popped out? स्टैक पर निम्नलिखित अनुक्रम में ऑपरेशन किए जाते हैं: PUSH (10), PUSH (20), POP, PUSH (10), PUSH (20), POP, POP, POP, PUSH (20), POP. पॉप आउट किए गए मान का अनुक्रम प्रदर्शित करें?	CO2	K3
e.	Illustrate the concept of Indexed Sequential search. अनुक्रमित अनुक्रमिक खोज की अवधारणा को स्पष्ट करें।	CO3	K4
f.	Explain the concept of NonLinear data structures with the help of example. उदाहरण की सहायता से नॉनलीनियर डेटा संरचना की अवधारणा को समझाइए।	CO4	K4
g.	Illustrate the significance of Threaded Binary Tree. थ्रेडेड बाइनरी ट्री के महत्व को स्पष्ट करें।	CO5	K4

SECTION B

2. Attempt any *three* of the following: 07 x 3 = 21

Q no.	Question	CO	Level
a.	An array ARR[-5…15, 10…20] stores elements in Row Major Wise with each element requiring 2 bytes of storage. Evaluate the address of ARR[10][15] when the base address is 2500. एक सरणी ARR[-5…15, 10…20] रो मेजर वाइज में तत्वों को संग्रहीत करती है, जिसमें प्रत्येक तत्व को 2 बाइट्स स्टोरेज की आवश्यकता होती है। जब आधार पता 2500 हो तो ARR[10][15] के पते का मूल्यांकन करें।	CO 1	K5
b.	If a stack is implemented using Linked List, then show the logical representation of nodes as a stack. Illustrate PUSH and POP operations of this type of stack. यदि लिंक्ड लिस्ट का उपयोग करके स्टैक को कार्यान्वित किया जाता है, तो स्टैक के रूप में नोड्स का तार्किक प्रतिनिधित्व दिखाएं। इस प्रकार के स्टैक के PUSH और POP संचालन को चित्रित करें।	CO 2	K4
c.	Construct a binary tree with the following traversals: **Inorder: B C A E G D H F I J** **Preorder: A B C D E G F H I J** निम्नलिखित ट्रैवर्सल के साथ एक बाइनरी ट्री बनाएं: इनऑर्डर: B C A E G D H F I J प्रीऑर्डर: A B C D E G F H I J	CO 3	K3
d.	Construct an AVL Tree by inserting following sequence of elements, starting with an empty tree: 71,41,91,56,60,30,40,80,50,55 एक खाली ट्री से शुरू करते हुए, तत्वों के निम्नलिखित अनुक्रम को सम्मिलित करके एक AVL ट्री का निर्माण करें: 71,41,91,56,60,30,40,80,50,55	CO 4	K3
e.	Explain the Depth-First Search (DFS) algorithm with the help of an example graph. Which data structure is used for DFS and BFS? एक उदाहरण ग्राफ की मदद से डेप्थ-फर्स्ट सर्च (DFS) एल्गोरिदम को समझाइए। DFS और BFS के लिए किस डेटा संरचना का उपयोग किया जाता है?	CO 5	K4

SECTION C

SECTION C

3. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Illustrate the following: a. Asymptotic notations b. Time Space Tradeoff c. ADT निम्नलिखित को स्पष्ट करें: a. असिम्टोटिक संकेतन b. समय स्थान व्यापार-बंद c. ADT	CO1	K4
b.	Illustrate how to represent the polynomial using linked list? Construct a C program to add two polynomials using linked list. लिंक्ड सूची का उपयोग करके बहुपद को कैसे दर्शाया जाए, इसका उदाहरण दें? लिंक्ड सूची का उपयोग करके दो बहुपदों को जोड़ने के लिए C प्रोग्राम बनाएँ।	CO1	K4

4. Attempt any *one* part of the following: 07 x 1 = 07

Q no.	Question	CO	Level
a.	Construct a Queue using a linked list? Include functions insert(), delete(), print().Explain how a **circular queue** overcomes the limitations of a simple queue. लिंक्ड सूची का उपयोग करके कतार का निर्माण करें? insert(), delete(), print() फ़ंक्शन शामिल करें। समझाएँ कि एक वृत्ताकार कतार एक साधारण कतार की सीमाओं को कैसे पार करती है।	CO2	K3
b.	Construct a C program to check whether the parentheses are balanced in an expression. किसी व्यंजक में कोष्ठक संतुलित हैं या नहीं, इसकी जाँच करने के लिए C प्रोग्राम का निर्माण करें।	CO2	K3

5. **Attempt any *one* part of the following:** **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Construct a Hash table that contains 10 buckets and uses linear probing to resolve collisions. The key values are integers and the hash function used is key % 10. Insert values 43, 165, 62, 123, 142 in the table. Show all the steps and find the location of the key value 142? एक हैश टेबल बनाएं जिसमें 10 बकेट हों और टकरावों को हल करने के लिए रैखिक जांच का उपयोग करें। कुंजी मान पूर्णांक हैं और उपयोग किया गया हैश फ़ंक्शन कुंजी % 10 है। तालिका में मान 43, 165, 62, 123, 142 डालें। सभी चरण दिखाएँ और कुंजी मान 142 का स्थान ज्ञात करें?	CO3	K3
b.	Demonstratemerge sort algorithm to sort the following elements in ascending order : 11, 16, 13, 11, 4, 12, 6, 7. What is the time and space complexity of merge sort? निम्नलिखित तत्वों को आरोही क्रम में सॉर्ट करने के लिए मर्ज सॉर्ट एल्गोरिदम का प्रदर्शन करें: 11, 16, 13, 11, 4, 12, 6, 7. मर्ज सॉर्ट की समय और स्थान जटिलता क्या है?	CO3	K3

6. **Attempt any *one* part of the following:** **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Demonstrate B-Tree? Construct a B-Tree of order 4 with the alphabets (letters) arrive in the sequence as follows: a g f b k d h m j e s i r x c l n t u p. बी-ट्री का प्रदर्शन करें? क्रम 4 का बी-ट्री बनाएं जिसमें अक्षर (अक्षर) इस प्रकार क्रम में हों: a g f b k d h m j e s i r x c l n t u p	CO4	K3
b.	Construct a **Binary Search Tree (BST)** using the following sequence of numbers: 50, 30, 70, 20, 40, 60, 80, 35, 45. Perform Inorder traversal of Tree. संख्याओं के निम्नलिखित अनुक्रम का उपयोग करके एक बाइनरी सर्च ट्री (BST) का निर्माण करें: 50, 30, 70, 20, 40, 60, 80, 35, 45. ट्री का इनऑर्डर ट्रैवर्सल करें।	CO4	K3

7. Attempt any *one* part of the following: **07 x 1 = 07**

Q no.	Question	CO	Level
a.	Explain two ways to represent a graph in memory and compare their advantages: (i) Adjacency matrix (ii) Adjacency List मेमोरी में ग्राफ को दर्शाने के दो तरीके बताएं और उनके फायदों की तुलना करें: (i) एडजेंसी मैट्रिक्स (ii) एडजेंसी लिस्ट	CO5	K4
b.	Explain the following graph terminologies with examples: (i) Graph (ii) Weighted Graph (iii) Degree of a Vertex (iv) Connected and Disconnected Graph (v) Cycle in a Graph (vi) Directed and Undirected Graph (vii) MST निम्नलिखित ग्राफ शब्दावली को उदाहरण सहित समझाइए: (i) ग्राफ (ii) भारित ग्राफ (iii) शीर्ष की डिग्री (iv) जुड़ा और डिस्कनेक्टेड ग्राफ (v) ग्राफ में चक्र (vi) निर्देशित और अनिर्देशित ग्राफ (vii) एमएसटी	CO5	K4

Thank You